Republic of Zambia

MINISTRY OF EDUCATION

CHEMISTRY

TEACHING MODULE

FORM 2

TERM 1 & TERM 2

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CHEMISTRY TEACHING MODULE FORM 2 [TERM 1 & TERM 2]

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Vision

Quality, life- long education for all which is accessible, inclusive and relevant to individual, national and global needs.

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Preface

The Chemistry Teaching Module for Form 2 is designed to support teachers in preparing and delivering competence-based lessons that promote a comprehensive understanding of Chemistry concepts. It aims to foster learners' deep appreciation of the role of Chemistry in everyday life and its applications across various fields of science and technology. The module assists teachers in helping learners develop analytical thinking, problem-solving abilities, and practical laboratory skills through a structured and progressive learning approach.

The Module emphasises hands-on activities and inquiry-based learning experiences that encourage learners to explore, experiment, think critically, and apply scientific reasoning in understanding chemical and physical phenomena while developing practical competencies. Furthermore, it guides teachers in creating a stimulating and supportive learning environment that enables learners to develop a profounder understanding of Chemistry.

Through carefully designed learning experiences, the module supports learners' intellectual and personal growth by cultivating curiosity, critical thinking, and practical skills, while preparing them for further education and potential careers in science, technology, and related fields. It is anticipated that this module will inspire learners to explore the fascinating world of Chemistry, develop curiosity about scientific phenomena, and appreciate the vital role Chemistry plays in shaping the future of science, technology, and society.

Kelvin Mambwe, (Dr)
Permanent Secretary- Educational Services
MINISTRY OF EDUCATION

Acknowledgement

The development of this **Chemistry Teaching Module** for **Form 2** was a collaborative effort. We would like to extend our sincere gratitude to the various Directorates, institutions, individuals, and subject associations that contributed to its successful development. We particularly thank the teachers and lecturers from Colleges of Education and the University of Zambia who participated in the development process. Their valuable insights, professional expertise, and constructive feedback were instrumental in shaping the content, structure, and overall direction of this module.

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Last but not least, we sincerely acknowledge the commitment and hard work of all staff at the Curriculum Development Centre, whose dedication ensured that this module became a reality.

Charles Ndakala, (Dr.)
Director - Curriculum Development
MINISTRY OF EDUCATION

How to use this Module

To effectively use this Chemistry Teaching Module for Form 2:

- Read and familiarise yourself with the module's content, learning activities, and assessment guides.
- Plan your lessons in advance, using the module's suggested teaching and learning activities.
- Use a variety of teaching methods, including demonstrations, discussions, group work, and hands-on experiments.
- Encourage active learning by asking open-ended questions, promoting critical thinking, and fostering problem-solving skills.
- Assess learners learning regularly, using the module's suggested assessment strategies and tools.
- Provide feedback and support to learners, helping them to identify areas for improvement and develop their skills.
- Integrate technology into your teaching, using multimedia resources and interactive simulations to enhance student engagement and understanding.
- Monitor student progress and adjust your teaching strategies as needed, to ensure that all students meet the learning objectives

By following these steps, you can effectively use this Chemistry Teaching Module for Form 1 to support your teaching and promote learning.

Introduction

Chemistry is a fundamental science that helps learners understand the composition, properties, and transformations of matter in the natural world. It enables learners to explain everyday phenomena and appreciate the role of science in improving the quality of life. As learners progress in secondary education, it is important that they develop a deeper understanding of key Chemistry concepts that form the foundation for further scientific learning. The **Chemistry Teaching Module for Form 2** is designed to support teachers in the effective delivery of the 2024 Chemistry Syllabus. The module provides guidance on the teaching and learning of important topics including Bonding, Chemical Reactions, Chemical Kinetics, and Redox Reactions. It includes topic and subtopic overviews, suggested teaching and learning materials, environmental set-up for practical activities, learner-centred learning activities, and assessment guidelines.

The module promotes learner-centred teaching approaches such as **inquiry-based learning, problem-solving activities, group discussions, and hands-on experiments**, which help learners actively engage with scientific concepts while developing critical thinking, analytical skills, and practical laboratory competences. Chemistry knowledge is essential for **national development**, as it supports key sectors of the economy such as **mining, agriculture, health, manufacturing, and environmental management**. By studying Chemistry, learners develop skills that enable them to understand scientific processes, solve real-life problems, and prepare for further studies and careers in science and technology. Overall, this module aims to support teachers in creating stimulating learning environments that encourage curiosity, experimentation, and an appreciation of the importance of Chemistry in everyday life and socio-economic development.

Suggested Teaching and Learning Materials

The study of Chemistry content requires hands-on experiences, visual aids, and interactive resources to strengthen deep understanding and appreciation. To support the teaching and learning of Chemistry in Form 1, the suggested teaching and learning materials are meant to:

- Enhance learner engagement and motivation in Chemistry
- Develop practical skills and laboratory techniques
- Promote critical thinking, problem-solving, and analytical skills
- Support differentiated instruction and inclusive learning

The suggested teaching and learning materials are either artificial or natural. By utilising these suggested teaching and learning materials, teachers can create a dynamic and supportive learning environment that promotes academic excellence, creativity, and scientific literacy in Chemistry for Form 1 learners.

Learning Environment Set Up

To create an effective learning environment for teaching and learning Chemistry and in deepening understanding of the concepts and application in real life context, the learning environment set-up aims to create a safe, inclusive, and engaging space.

- **Natural Environment**: A natural learning environment is a setting where learners explore and learn naturally, often without explicit instruction or formal teaching, such as in school surroundings
- **Man-Made Environment:** Man-made learning environments are intentionally designed safe spaces, such as classrooms, laboratories, and libraries, designed for formal instruction, hands-on activities, and games and songs
- **Technological Learning Environment:** Access educational apps, games, and software for learning, including game-based platforms, virtual platforms, and simulations, to engage learners and promote learning.

Safety in the Learning Environment

Safety in the learning environment is a requirement in learning Chemistry, as learners are exposed to potential hazards during hands-on experiments. Collaboration between teachers, and learners, is important to create a responsible learning environment that promotes scientific inquiry. Guidelines for maintaining a safe learning environment include laboratory safety rules (protocols), risk identification, personal protective equipment, emergency response plans, chemical handling, storage, and disposal, and learner responsibilities. Prioritizing safety minimizes risks, prevents accidents, and ensures a positive learning experience.

Suggested Teaching Methodology

The effective teaching methodologies in STEM Chemistry include:

- **Conducting experiments**: Demonstrate key principles and encourage curiosity among learners.
- **Collaborative learning**: Pair learners to work together, promoting peer-to-peer teaching, discussion, and problem-solving.
- **Conceptual learning:** Connect chemical concepts to everyday life, industry, or current events, making learning relevant and meaningful.
- **Differentiated instructions:** Tailor teaching to meet diverse learning styles, abilities, and interests of different learners.
- **Feedback and reflection:** Encourage learners to reflect on their learning, providing constructive feedback to guide improvement.
- **Inquiry-based learning**: Encourage learners to explore, investigate, and discover chemical concepts through hands-on experiments and activities.
- **Integration of technology:** Use digital tools, simulations, and visualizations to enhance engagement, understanding, and analysis.

- **Problem-based learning**: Present real-world problems or case studies or scenarios, requiring learners to apply chemical knowledge to develop solutions.
- **Project -based learning:** Assign open-ended projects, allowing learners to design, conduct, and present research or applications of chemical concepts.

By implementing these methodologies, a teacher can create an engaging, inclusive, and effective STEM Chemistry learning environment.

Learning Activities

Learning activities are intentional educational experiences aimed at promoting learning, engagement, and achievement among learners. Facilitated by teachers, they help acquire new knowledge, skills, attitudes, and behaviour change. To create an inclusive environment, teachers should use a "**hook**" or problem posing or scenario or key question or case studies to introduce new learning activities in an interactive and interesting way.

Time Allocation

The standard minimum learner-teacher contact time for Chemistry at secondary school level is four (4) hours per week, translating to Six (6) periods with at least two double periods per week. The duration for a single period is 40 minutes. The contact time at Ordinary Secondary school level is planned in such a way as to give ample time for practical activities.

Icons used in this Module

This module utilises icons as visual symbols or graphics to represent instructions, enhancing the learning experience for learners. Icons categorise and organise instructions, making navigation easier for teachers.

 Learners Activities	 Assessment Guide	 Key Terms	 Skills Developed	 Introduction
 **Discussion**	 Key Learning Points	 Tips	 General Competences	 Learners' Tasks

Assessment

- Formative Assessments: To monitor **learner** progress, identify areas of improvement, and adjust instruction to meet **learner** needs.
- Quizzes: Regular quizzes to assess **learner**s' understanding of concepts.
- Tests: Periodic tests to evaluate **learner**s' knowledge and application of Chemistry concepts.
- Class Discussions: Observing **learner**s" participation and engagement in class discussions.
- Laboratory Reports: Evaluating **learner**s' lab**oratory** reports for accuracy, completeness, and understanding.
- Group Work: Assessing **learner**s' ability to work collaboratively and contribute to group tasks.
- Projects**: Evaluating learner**s' **ability to design, conduct, and present a Chemistry project.**
- Presentations: Assessing learners' ability to communicate Chemistry concepts and ideas effectively

Summative Assessments: To evaluate learner learning at the end of a lesson, sub-topic, topic or term, and to provide a comprehensive picture of learner achievement.

- **Unit Tests**: Comprehensive tests to evaluate learners' understanding of Chemistry concepts at the end of each unit.
- **Mock Exams**: Comprehensive exams to evaluate learners' overall understanding of Chemistry concepts at the end of the semester.
- **Practical Tasks**: Assessing learners' laboratory skills and techniques through practical exams.
- **Project-Based Assessments**: Evaluating learners' ability to design, conduct, and present a Chemistry project.

Key Competences

KEY COMPETENCES	DESCRIPTORS
Analytical Thinking	• Identify patterns • Compile data, create mental images and address issues • Evaluate solutions
Collaboration	• Solving puzzle in groups • Play with peers to build relationships • Participate in and express themselves through play activities
Communication	• Use mathematical/scientific language in different situations • Express oneself using different media and symbols • Ask for feedback

KEY COMPETENCES	DESCRIPTORS
Critical Thinking	• Ask and answer simple questions • Classify objects according to their attributes • Manipulate different objects • Solve simple problems in life • Match different things according attributes • Arrange objects according to attributes • Compare similarities or differences between objects • Explore the environment • Differentiate good from bad • Recognize and name items in the environment
Environmental Sustainability	• Dispose trash in the designated place. • Adhere to best practices in environmental management. • Identify a clean environment. • Identify types of waste in local environment
Problem Solving	• Make connections/link with the inner world or social environment • Use numeracy patterns and relations to solve problems • Manipulate numbers, shapes and symbols to complete a task

TERM 1

TOPIC 2.1: BONDING

Introduction:

Atoms of most elements are naturally unstable. In their tendency to become stable, they interact with each other through the process of bonding. The bonding process involves the sharing or transfer of electrons between atoms in order for them to obtain stable electron configurations. In the process, molecules or ions are formed. Bonding is the formation of attraction forces between atoms, or ions, or molecules. This process leads to the formation of various types of chemical compounds with diverse structures and properties. This topic will look at the various types of bonding and how they take place.

General Competences:

- Collaboration
- Communication
- Analytical Thinking
- Environmental Sustainability
- Problem Solving

Sub-Topic 2.1.1: Chemical Bonding

Introduction

Molecules and compounds are formed by the process called chemical bonding. Chemical bonding takes place in different ways between atoms of the same element and / or atoms of different elements.

Specific Competence:

2.1.1.1 Demonstrate understanding of chemical bonding

Key Term:

- **Bonding:** The combining of atoms or elements to form molecules or compounds

Learning Activities:

The following activity will help learners understand what chemical bonding is, and analyse the different types of bonding.

Activity 1.0: Describing chemical bonding and analysing the types of bonding

The following activity is designed to help learners develop an understanding of what chemical bonding is and analyse the different types of bonding. In this activity, learners will work in groups to research and find solutions to given questions.

HOOK: In some cultures, each man is expected to marry one woman while in other cultures a man can marry two or more women. Would such combinations be possible among atoms? Discuss.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Computers connected to internet, science dictionary, text books, charts, markers, work cards, bostik, name and chemical formula tags for oxygen molecule (O_2), sodium chloride (NaCl), magnesium oxide (MgO), water (H_2O), hydrogen chloride (HCl), copper metal (Cu), aluminium metal (Al)

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with charts, markers, bostik, tags with the name and chemical formula of the following: oxygen (O_2), sodium chloride (NaCl), hydrogen chloride (HCl), copper (Cu), aluminium (Al) or any other metals
- Ask learners to work in groups and research using the following questions on work cards:
    - What is chemical bonding?
    - What is a bond?
    - Examine the substances on the tags and identify the types of elements bonded together in each substance
    - Categorise the substances based on the type of elements bonded together in them
    - Discuss, identify and name the type of bonding between the elements in the identified categories of the substances.
- Ask groups to present their findings to the class
- Consolidate presentations from learners

Learners' Activities:

- Working in groups using the provided materials
- Discussing and answering the given questions on the work cards
- Making presentations to the class

Assessment Guide:

- Ask learners the different types of bonding they come up with
- Check if learners are:
  - describing bonding correctly
  - identifying the bonded elements correctly
  - categorising the substances based on their bonding correctly

Key Learning Points:

- Bonding is the process by which atoms combine chemically to form molecules or compounds.
- There are different types of bonding which include
  - Ionic bonding which takes place between a metal and a non-metal. It involves the transfer of electrons from a metal to a non-metal leading to formation of oppositely charged ions.
  - Covalent bonding which takes place between atoms of non-metals leading to the formation of molecules.
  - Metallic bonding which occurs within a metal leading to the formation of a metallic lattice.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying

Expected Standard:

- Understanding of chemical bonding demonstrated correctly.

Sub-Topic 2.1.2: Ions

Introduction

Atoms of elements are naturally neutral. This neutrality arises from the fact that the number of protons is equal to the number of electrons, which makes the number of positive and negative charges equal in an atom. However, the number of electrons and protons in an atom can differ. This difference leads to the formation of ions.

Specific Competence:

2.1.2.1 Demonstrate understanding of ion formation in bonding**.**

Key Terms:

- **Ion**: A charged particle which is formed when an atom loses or gains an electron(s).
- **Polyatomic ion**: A group of atoms with a net charge
- **Cation**: A positively charged ion
- **Anion**: A negatively charged ion.
- **Radica**l: An atom or group of atoms with unused valencies
- **Valency**: The combining power of an element
- **Valence**: The number of electrons in the outer most shell of an atom

Learning Activities:

Activity 1.0: Describing an ion and the formation of monoatomic ions

The following activity is designed to help learners develop an understanding of what ions are, the types of ions and how they are formed.

HOOK: Imagine two learners in class. One has extras pencils while the other one has few. The learner with many pencils gives one away while the other receives one. The learner who gave away a pencil has one less and the one who receives a pencil has an extra one. If you were to assign charges to the learners, which charges would you assign to each learner? Discuss

Suggested Teaching and Learning Materials

- **Natural Materials**:
- **Artificial Materials:** Computers connected to internet, text books, charts, markers of different colours, work cards and bostik

Learning Environment Set-up:

- **Natural Environment**: School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide each group with the materials: charts, markers, bostik
- Ask learners to draw electronic configuration diagrams for the atoms of the elements, sodium, magnesium, fluorine and oxygen
- Ask learners the following questions:
    - Which of the atoms will lose electrons and which ones will gain electrons in order to become stable?
    - For those that will lose, how many electrons will they lose each?
    - For those that will gain, how many electrons will they gain each?
    - What will be the net charge on each atom after losing or gaining electrons?
    - Which type of elements lose electrons, and what type of ions do they form
    - Which type of elements gain electrons, and what type of ions do they form
- Ask learners to come up with their own description of an ion
- Ask groups to present their findings to the class
- Consolidate learners' presentations

Learners' Activities:

- Drawing electronic configuration diagrams for the atoms of the elements, sodium, magnesium, fluorine and oxygen
- Identifying the atoms that will lose electrons and the ones that will gain electrons in order to become stable?
- Working out how many electrons each will lose or gain in order to become stable
- Working out the net charge on each atom after losing or gaining electrons
- Identifying the type of elements that lose electrons, and the type of ions they form
- Identifying the type of elements that gain electrons, and the type of ions they form
- Describing an ion in own words

Assessment Guide:

- Check if learners are:
    - drawing electronic configurations correctly
    - identifying the elements that lose or gain electrons correctly
    - working out the net charges on ions correctly

- describing the term ion correctly
- collaborating as they work in groups

Key Learning Points:

- Atoms of metals lose electrons while atoms of non-metals gain electrons in order to become stable. When this happens, ions are formed
- When metals lose electrons, they form positively charged ions called cations
- When non-metals gain electrons, they form negatively charged ions called anions
- A monoatomic ion is an ion formed from a single atom when it gains or loses electrons.
- Radicals are usually referred to as polyatomic ions
- When an atom loses one or more electrons, it becomes positively charged. This is because the number of protons (positive charges) is now greater than the number of electrons (negative charges).
- When an atom gains one or more electrons, it becomes negatively charged. This is because the number of electrons (negative charges) is now greater than the number of protons (positive charges).

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 2.0: Relating the charges of ions and radicals to their valencies

The following activity is designed to help learners to be able to relate the charges on the ions and radicals to their valencies.

HOOK: Analyse the two ions shown: Mg^{2+} and SO_4^{2-}
What do the charges on the ions reveal to you?

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Computers connected to internet, text books, flip charts, markers, work cards, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with the materials: charts, markers, Bostik
- Ask learners to work in groups to research, discuss and do the following:
    - Describe the terms
        - radical,
        - valence of elements
        - valency of elements
    - Research on the valencies of elements and radicals
    - Use electronic configurations of sodium, calcium, chlorine and oxygen to determine their valencies
    - Discuss how sodium, calcium, chlorine and oxygen form ions, noting the number of electrons they lose or gain
    - Determine the charges on their ions
    - Write the formulae of their ions
    - Relate the number of charges on the ions to their valencies
    - Research on radicals and their valencies
    - Relate the number of charges on chloride, ammonium and sulphate radicals to their valencies
    - Present their findings to the class

Learners' Activities:

- Describing the terms radical, and valence and valency of elements
- Researching on the valencies of elements and radicals
- Determining valencies of sodium, calcium, chlorine and oxygen using their electronic configurations
- Discussing how sodium, calcium, chlorine and oxygen form ions, noting the number of electrons they lose or gain
- Determining the charges on their ions
- Writing the formulae of their ions
- Relating the number of charges on their ions to their valencies
- Researching on radicals and their valencies

- Relating the number of charges on chloride, ammonium and sulphate radicals to their valencies
- Presenting findings to the class

Assessment Guide:

- Check if the learners are
  - describing the terms correctly
  - determining the valencies correctly
  - writing the formulae of ions correctly
  - working out the net charges on ions correctly
  - drawing the relationship between number of charges on ions/radicals and their valencies correctly

Key Learning Points:

- The number of charges (positive or negative) on the ions is equal to the number of electrons lost or gained by an element. This number is equal to the valency of the element. Similarly, the number of charges on a radical is equal to the valency of the radical.
- While valence refers to the number of electrons in the outer most shell of an atom, valency is the combining power of an element. It is a small whole number that signifies the number of bonds an element is able to form in bonding. The valency number is equal to the number of electrons an element loses or gains to become stable. The ions formed carry an equal number of charges. The valency of an element is determined by the number of valence electrons.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standard:

- Understanding of ion formation in bonding demonstrated accordingly

Sub-Topic 2.1.3: Chemical Formulae

Introduction

A chemical formula is a shorthand representation of a molecule or a compound. It is a way of representing an element or a compound using chemical symbols of the elements and numbers. It shows the elements that make up a molecule or a compound in their simplest form. Chemical formulae show the ratio of atoms in a molecule of an element or compound or a formula unit. For example, the formula of a hydrogen molecule is H_2, of water is H_2O, and calcium chloride is $CaCl_2$.

Specific Competences:

2.1.3.1 Deduce chemical formulae of compounds

2.1.3.2 Demonstrate understanding of chemical compounds

Key Terms:

- **Chemical formula**: A shorthand representation of a molecule or compound
- **Binary compound**: A compound made up of two elements
- **Ternary compound**: A compound made up of three elements

Learning Activities:

Activity 1.0: Assigning of valencies to elements and radicals

In this activity, learners will engage in research on the concept of valencies of elements and an activity on assigning valencies to elements and radicals. Learners will further relate the valencies of elements and radicals to the number of bonds they can form.

HOOK: The salt we use in daily life in our meals is formed when sodium and chlorine combine. They combine in a fixed ratio. Discuss what causes this fixed ratio combination.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Computers connected to internet text books, flip charts, markers, work cards, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity

- Put learners in small groups
- Provide each group with the materials
- Ask learners to work in groups to research, discuss and do the following:
    - Research on the valencies of elements and radicals
    - Create a table for the first 20 elements with columns headed: name of element, chemical symbol, atomic number, electronic configuration, valence and number of bonds it can form
    - Use electronic configurations to determine and assign valencies to elements
    - Create a similar table for radicals with headings: name of radical, chemical symbol/formula, valency and number of bonds it can form
    - Present their findings to the class

Learners' Activities:

- Researching on the valencies of elements and radicals
- Creating a table for the first 20 elements with columns headed: name of element, chemical symbol, atomic number, electronic configuration, valency and number of bonds it can form
- Determining and assigning valencies to elements using electronic configurations
- Creating a similar table for radicals with headings: name of radical, chemical symbol/formula, valency and number of bonds it can form
- Presenting findings to the class

Sample table to be developed by learners

SN	Name of element	Chemical symbol	Atomic number	Electronic configuration	Valency	Number of bonds it can form
1.						
2.						

Assessment Guide:

- Check if the learners are deducing valencies correctly

Key Learning Points:

- Each element has a valency which is represented by a small whole number. This valency number is referred to as the combining power which an element or radical uses to combine with other elements. The valency number also represents the number of bonds that element or radical forms in combining with others.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Identifying

Activity 2.0: Formulating chemical formulae of compounds

The chemical formulae of compounds can be formulated using valencies of elements or charges on ions. Learners will engage in two activities to formulate chemical formulae of compounds.

Activity 2.1: Formulating chemical formulae of compounds using valencies

In this activity, learners will work in groups to construct chemical formulae of compounds using valencies. This will involve combining of two species that is, either two elements, or an element and a radical, or two radicals.

HOOK: Water is one of the common and very useful compounds in life. The chemical formula of water is presented as H_2O. Explain how this formula is arrived at?

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Flip charts, markers, work cards with names of elements and radicals, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide each group with the materials: flip charts, Bostic, work cards with names of elements and radicals written on them

Work cards

Potassium	Magnesium	Oxygen	Carbonate	Fluorine
______	______	______	______	______

Aluminium	Phosphate	Hydrogen sulphate	Hydroxide

- Ask learners to work in groups to do the following:
    1. Pick any two elements or radicals that can form a compound from the work cards provided and stick on the flip chart (Note that a card can be used more than once)
    2. Write the chemical formula of the compound that can be formed from the elements or radicals on the flip charts, below the name of the compound
    3. Repeat steps 1 and 2 to formulate as many chemical formulae as possible
    4. Present work on flip charts to the class
- Consolidate on the presentations by the groups

Learners' Activities:

1. Picking any two elements or radicals that can form a compound from the work cards provided and sticking on the flip chart
2. Writing the chemical formula of the compound that can be formed from the elements or radicals on the flip charts, below the name of the compound
3. Repeating steps 1 and 2 to formulate as many chemical formulae as possible
4. Presenting work on flip charts to the class

Assessment Guide:

- Check if the learners are formulating the chemical formulae of compounds correctly

Key Learning Points:

- Writing of chemical formulae of compounds involves the following steps:
    - identifying the combining elements and /or radicals
    - writing the symbols of the combining elements/radicals starting with the metal or a more electropositive element
    - assigning valencies, and then swapping or interchanging the valencies
    - writing the chemical formula with the swapped valencies written as subscripts
    - reducing the valencies to their lowest terms if need be

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Identifying

Activity 2.2: Formulating chemical formulae of compounds using charges on the ions

In this activity, learners will use charges on the ions to construct chemical formulae of compounds.

HOOK: Elements form different types of ions with varying number of charges. Discuss the information that nature of ions and number of charges on ions or radicals reveal about the elements.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Flip charts, bostik, work cards with chemical formulae of different cations and anions written on them

Learning Environment Set-up:

- **Natural Environment**: School surroundings
- **Artificial Environment:** Classroom/ Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide each group with the materials: flip charts, Bostic, work cards with chemical formulae of different cations and anions written on them

Work cards

Na^+	Ca^{2+}	OH^-	NO_3^-	SO_4^{2-}
O^{2-}	Al^{3+}	Cl^-	NH_4^+	

- Ask learners to work in groups to do the following:
    1. Pick any two ions that can form a compound from the work cards provided (Note that a card can be used more than once)
    2. Analyse the charges on the ions
    3. Write the chemical formula of the compound that can be formed from the ions on the flip charts
    4. Repeat steps 1, 2 and 3 to formulate as many chemical formulae as possible
    5. Present their work on flip charts to the class
- Consolidate on the presentations by the groups

Learners' Activities:

1. Picking any two ions that can form a compound from the work cards provided

2. Analysing the charges on the ions
3. Writing the chemical formula of the compound that can be formed from the ions on the flip charts
4. Repeating steps 1, 2 and 3 to formulate as many chemical formulae as possible
5. Presenting work on flip charts to the class

Assessment Guide:

- Check if the learners are formulating the chemical formulae of compounds correctly

Key Learning Points:

- Writing of chemical formulae of compounds using charges of ions involves the following steps:
    - identifying the combining ions
    - noting the charges of the combining ions
    - deducing the valencies of the combining ions from the charges
    - writing the chemical formulae of the ions starting with the cation followed by the anion
    - swapping or interchanging the valencies and writing them as subscripts in the formula
    - reducing the valencies to their lowest terms if need be

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 3.0: Naming of chemical compounds

In this activity, learners will explore by research the system of naming binary and ternary compounds.

HOOK: Usually, the name of a person has two parts. Part of the information the name reveals is the parent’s name of that person. How does this apply to names of compounds?

Suggested Teaching and Learning Materials:

- **Natural Materials:**

- **Artificial Materials:** Computers connected to internet, text book, flip charts, markers, bostik, work cards with chemical formulae of binary and ternary compounds

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide each group with the materials: text books, flip charts, markers, bostik, computers with internet, work cards with chemical formulae of binary and ternary compounds

Work cards

$CaCl_2$	Na_2O	$FeCl_3$	$MgSO_4$
CuO	$ZnCO_3$	KNO_3	$(NH_4)_2SO_4$

- Ask learners to work in groups to do the following:
    - research on the naming of compounds
    - compile the rules for naming binary and ternary compounds
    - use research findings to assign names to the compounds provided on the work cards
    - Present research findings and the names to the class
- Consolidate on the presentations by the groups

Learners' Activities:

- Researching on the naming of compounds
- Compiling the rules for naming binary and ternary compounds
- Assigning names to the compounds provided on the work cards using research findings
- Presenting research findings and the names to the class

Assessment Guide:

- Check if the learners are:
    - applying the rules of naming compounds correctly
    - correctly assigning names to the chemical formulae of compounds

Key Learning Points:

- Binary compounds are compounds of two elements which can be either ionic (e.g. NaCl, CaO) or covalent (e.g. NO_2, N_2O_4, N_2O, SO_3) in nature.
- Naming binary compounds depends on whether the compound is ionic or covalent. In ionic binary compounds, the name starts with the element's name for the cation followed

by the name of the anion with an "ide" suffix e.g. sodium chloride and calcium oxide. For covalent compounds, prefixes such as mono- for 1, di- for 2, tri- for 3, tetra- for 4 are used to show the number of atoms. If the prefix is mono in the first element, it is dropped while the second element ends with the "ide" suffix e.g. nitrogen dioxide (NO_2), dinitrogen tetroxide (N_2O_4), Sulphur trioxide (SO_3), dinitrogen monoxide (N_2O).

- For cations with variable charges, e.g. Fe^{2+} and Fe^{3+}, roman numerals in brackets are used in the name to show the charge, for example, iron (II) chloride for $FeCl_2$ and iron (III) oxide for Fe_2O_3
- Ternary compounds are compounds of three different elements, usually a metal cation and a polyatomic anion (radical) e.g. $CaCO_3$. Naming ternary compounds is done by stating the metal or cation first followed by the polyatomic anion (radical) e.g. Sodium sulphate for Na_2SO_4, iron (II) nitrate for $Fe(NO_3)_2$

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standards:

- Chemical formulae of compounds deduced accordingly
- Understanding of chemical compounds demonstrated accordingly

Sub-Topic 2.1.4: Ionic Bonding

Introduction

Ionic bonding involves the transfer of electrons from a metal to a non-metal. It involves the loss and gain of electrons between metallic and non-metallic atoms. A metal loses electrons while a non-metal gains electrons.

Specific Competences:

2.1.4.1 Demonstrate understanding of ionic bonding

2.1.4.2 Apply the uses of ionic compounds in our everyday life

Key Terms:

- **Ionic bond:** Electrostatic force of attraction between cations and anions
- **Cations**: Cations are positively charged ions. They are particles formed when atoms of metals lose one or more electron(s)
- **Anions**: Anions are negatively charged ions. They are particles formed when atoms of non-metals gain one or more electron(s)

Learning Activities:

There are four learning activities designed to help learners demonstrate understanding of ionic bonding as well as apply the uses of ionic compounds in everyday life. The first activity focuses on describing an ionic bond, second activity focusses on describing the formation of ionic (electrovalent) compounds, third activity focuses on explaining the properties of ionic compounds and the fourth activity is on exploring the uses of ionic compounds.

Activity 1: Describing an ionic bond

In this activity, learners will work in small groups to describe an ionic bond.

HOOK: Imagine you're an engineer fixing an aging bridge. Your team notices that steel beams rust from road salt (NaCl) eating through the beams. Why does the salt's sodium and chlorine "lock" onto iron so tightly, forming a tough ionic shield?

Suggested Teaching and Learning Materials:

- **Natural Materials**: -
- **Artificial Materials**: Textbooks, Science dictionaries, flip charts, markers, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put the learners in small groups
- Distribute the suggested or any improvised materials
- Ask learners to:
    - respond to the hook
    - use digital devices, science dictionary or textbooks to search for the description of an ionic bond
    - further search for more information about an ionic bond
    - discuss findings with classmates for peer review
    - write the description of an ionic bond in exercise books
    - make presentations to the class

Learners' Activities:

- Providing responses to the hook
- Searching for the description of an ionic bond using digital devices, science dictionary or textbooks
- Searching for more information about an ionic bond
- Discussing findings with classmates for peer review
- Writing the description of an ionic bond in the exercise books
- Making presentations to the class

Assessment Guide:

- Observe learners' participation during the activity
- Check if learners are describing ionic bond correctly
- Check the correctness of the extra information on an ionic bond

Key Learning Points:

- An ionic bond is the electrostatic force of attraction between cations and anions. It is a type of chemical bond formed between two atoms through the complete transfer of one or more electrons from one atom to another, resulting in the formation of oppositely charged ions that attract each other. Ionic bonds occur between metals and non-metals.

Skills Developed:

- Analytical Thinking
- Collaborating

- Communicating

Activity 2: Describing the formation of ionic (*electrovalent*) compounds

In this activity, learners will work in small groups to describe how ionic compounds such as sodium chloride, magnesium oxide and calcium chloride are formed.

HOOK: An Engineer tests magnesium oxide (MgO) as a road salt alternative to $NaCl$. Trucks passing spread MgO, but Mg instantly bond with oxygen in the mix, creating a super-stable compound that resists washouts?

Suggested Teaching and Learning Materials:

- **Natural Materials**: Table salt
- **Artificial Materials**: Flip charts, markers, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with the suggested or any improvised appropriate materials
- Ask learners to describe how the following compounds are formed:
    (a) Sodium chloride
    (b) Magnesium oxide
    (c) Calcium chloride
- Ask learners to draw dot and cross electronic configuration diagrams to show the bonding in sodium chloride, magnesium oxide and calcium chloride
- Ask learners to make presentations

Learners' Activities:

- Discussing the hook in relation to ionic bonding
- Describing how the following compounds are formed:
    (a) Sodium chloride
    (b) Magnesium oxide
    (c) Calcium chloride
- Drawing dot and cross electronic configuration diagrams to show the bonding in sodium chloride, magnesium oxide and calcium chloride

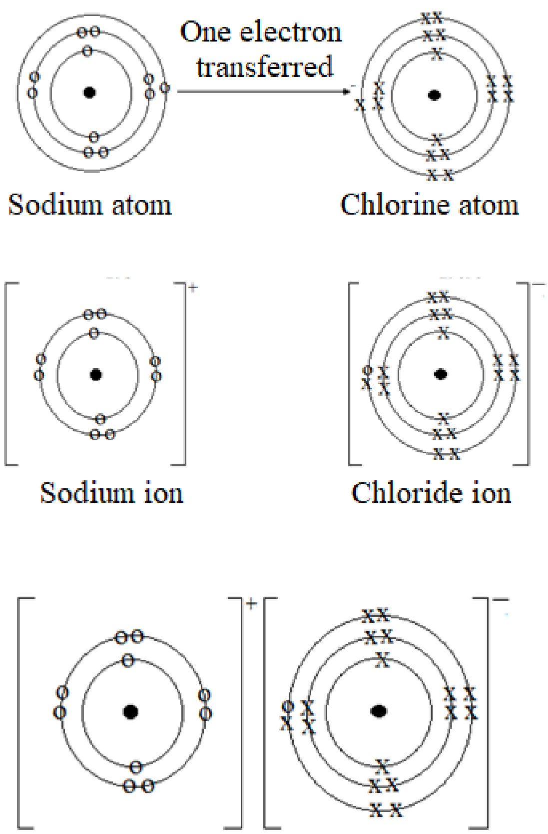
- Making presentations

Assessment Guide:

- Check if learners are:
    - describing the formation ionic compounds correctly
    - drawing correct and appropriate dot and cross electronic configuration diagrams

Key Learning Points:

- Ionic compounds consist of positive and negatively ions. Metals such as sodium, magnesium and calcium lose electrons to form positively charged cations. Non-metals such as oxygen and chlorine gain electrons to form negatively charged anions. The ions attract each other to form a neutral compound and the ionic bond is formed between the oppositely charged ions. The electrostatic attraction between the oppositely charged ions forms the ionic bond.



NaCl

Skills Developed:

- Analytical Thinking

- Collaborating
- Communicating

Activity 3: Explaining the properties of ionic compounds

In this activity, learners will work in small groups to explain the properties of ionic compounds.

HOOK: A learner in a rural area needs to make a working simple electrical circuit using materials available in the kitchen. There is salt, sugar, and sand. The learner mixes each of the materials with water and want to know which solution will allow electricity to flow and light a small bulb.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Flip charts, markers, bostik, work sheet showing the properties of some selected ionic compounds.

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide each group with a worksheet and ask them to complete the last column
- Ask learners to:
    - explain the trend in melting and boiling points of ionic compounds.
    - explain why ionic compounds do not conduct electricity when solid but conduct when molten or in solution
    - explain the higher density of ionic compounds
    - explain why ionic compounds are solids at r.t.p
    - discuss and share their findings with classmates.

Learners' Activities:

- Discussing the hook.
- Completing the following worksheet by writing conducts or does not conduct:

Compound	**Melting Point/^{0}C**	**Boiling Point/^{0}C**	**Density g/cm^3**	**Conductivity**	
				Solid	**Molten**
Sodium	801	1, 413	1.16		

chloride					
Magnesium oxide	2, 852	3, 600	1.58		

- Explaining
    - the trend in melting and boiling points of ionic compounds.
    - why ionic compounds do not conduct electricity when solid but conduct when molten or in solution
    - the higher density of ionic compounds
    - why ionic compounds are solids at r.t.p
- Discussing and making presentations to the class

Assessment Guide:

- Check if learners are explaining the properties of ionic compounds correctly

Key Learning Points:

- Ionic compounds are made up of positively and negatively charged ions. This results in strong electrostatic forces of attraction between the ions leading to:
    - high melting and boiling points. The strong electrostatic forces of attraction require large amount of energy to break, resulting in high melting and boiling points.
    - high densities. This is because their ions are closely packed together in a regular lattice.
    - solids state at room temperature and pressure and being non-volatile.
- They are soluble in water but insoluble in organic solvents such as ethanol and petrol.
- They conduct electricity in aqueous solution or molten state because the ions are free to move. They do not conduct electricity when in solid form because the ions are in fixed positions

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating

Activity 4: Exploring the uses of ionic compounds

In this activity, learners will work in small groups to explore the uses of ionic compounds.

HOOK: A farmer notices that crops in one part of his field are not growing well. The plants are small and yellowish. An agricultural officer advises the farmer to spread lime (CaO) on the soil before planting again.
Why did the agricultural officer recommend adding lime to the soil? Discuss

Suggested Teaching and Learning Materials:

- **Natural Materials**: Table salt
- **Artificial Materials**: Soap, batteries, water softeners, road de-icers, toothpaste, baking soda, fertilizers, lime, fire works

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in pairs or small groups and provide the materials
- Ask learners to:
    - discuss the uses of each of the materials provided
    - identify ionic compounds in the materials provided
    - share more examples of ionic compounds they use in everyday life
- Move around to assist learners and facilitate discussions
- Ask each group to present findings

Learners' Activities:

- Providing responses to the hook
- Discussing the uses of each of the materials provided
- Identifying ionic compounds in the materials provided
- sharing more examples of ionic compounds they use in everyday life
- Presenting the findings

Assessment Guide:

- Observe learner participation during the activity
- Assess learners with questions on the uses of different ionic compounds

Key Learning Points:

Ionic compounds have several uses both in everyday life and in industry. The following are some of the ionic compounds and their uses:

- Calcium oxide is used in the lining of furnaces. It is used as a refractory material. It is also used in agriculture to neutralize acidic soils.

- Sodium chloride is used as table salt for food seasoning and as a preservative.
- Calcium chloride is used as a drying agent in laboratories, ship containers and in the manufacturing of acetylene gas. It has a drying capacity of about 98%.
- Barium chloride is used in fireworks since it gives a green-colored explosion.
- Iron (II) sulphate is used in iron tablets given to pregnant women and anaemia patients.
- Calcium fluoride is the major compound in toothpaste.
- Lithium chloride is used in batteries as an electrolyte.
- Salts are generally used as Road De-icers

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating
- Observing
- Problem solving

Expected Standards:

- Understanding of ionic bonding demonstrated correctly.
- The uses of ionic compounds in our everyday life applied accordingly.

Sub-Topic 2.1.5: Covalent Bonding

Introduction

Covalent bonding involves the sharing of the outer most electrons between non-metallic atoms to form molecules. A molecule is a group of covalently bonded atoms which can exist independently. As a result of sharing electrons, each non-metal acquires a completely filled outer most shell. Covalent bonding leads to the formation of many compounds that have wide uses in everyday life and industry.

Specific Competences:

2.1.5.1 Demonstrate understanding of covalent bonding

2.1.5.2 Apply the uses of covalent compounds in everyday life

Key Terms:

- **Covalent bond**: A shared pair of electrons
- **Molecule**: A group of covalently bonded atoms which can exist independently.
- **Volatility**: The tendency of a substance to change state from liquid to gas

Learning Activities

There are five learning activities designed to help learners demonstrate understanding of covalent bonding as well as apply the uses of covalent compounds in our everyday life.

Activity 1: Describing a covalent bond

In this activity, learners will work in small groups to describe a covalent bond.

HOOK: Non-metals that do not have stable electronic configurations attain stability among themselves without gaining or losing electrons. In covalent bonding, atoms share electrons. How do they acquire stability among themselves?

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Computers connected to internet, textbooks, Science dictionaries, flip charts, markers, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity

- Put the learners in small groups
- Distribute the suggested or any improvised materials
- Ask learners to:
    - use digital devices, science dictionary or textbooks to search for the description of a covalent bond
    - further search for more information about a covalent bond
    - discuss findings with classmates for peer review
    - write the description of a covalent bond in exercise books
    - make presentations to the class
- Consolidate presentations by the learners

Learners' Activities

- Providing responses to the hook
- Searching for the description of a covalent bond using digital devices, science dictionary or textbooks
- Searching for more information about a covalent bond
- Discussing findings with classmates for peer review
- Writing the description of a covalent bond in the exercise books
- Making presentations to the class

Assessment Guide:

- Observe learners' participation during the activity
- Check if learners are describing a covalent bond correctly
- Check the correctness of the extra information on a covalent bond

Key Learning Points:

- A covalent bond is a shared pair of electrons between non-metals. Atoms are held by the attraction between their positive nuclei and the shared electrons.

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating
- Observing

Activity 2: Justifying the formation of single, double and triple bonds between atoms

In this activity, learners will work in small groups to describe how covalent compounds such as hydrogen, oxygen and nitrogen molecules are formed.

HOOK: Oxygen gas is a life supporting gas needed for aerobic respiration in living things. The two atoms in the molecule of oxygen gas, O_2, are stable. Justify the presence of the double bond (O = O) in the molecule.

Suggested Teaching and Learning Materials:

- **Natural Materials**: Table salt
- **Artificial Materials**: Flip charts, markers, bostik, medium balls of clay or plasticine, tooth picks, pens, glue sticks, paper

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with the suggested or any improvised appropriate materials
- Ask learners to do the following tasks:

  (a) Hydrogen
    - Use two small balls of clay or plasticine to represent two hydrogen atoms
    - Insert one tooth pick between them to represent one pair of shared electrons.
    - Using a paper, pen and glue stick, label the toothpick as a single covalent bond

  (b) Oxygen
    - Use two medium balls of clay or plasticine to represent two oxygen atoms
    - Insert two tooth picks between them to represent two pairs of shared electrons
    - Label it as a double covalent bond

  (c) Nitrogen
    - Use two medium balls of clay or plasticine to represent two nitrogen atoms
    - Insert three tooth picks between them to represent three pairs of shared electrons
    - Label it as a triple covalent bond
- Ask learners to:
  - explain in terms of electrons how the bonds in the models of hydrogen, oxygen and nitrogen are formed
  - draw the electron diagrams showing the bonding in the models they constructed

    - make presentations
    - display their models on the classroom notice board

Learners' Activities:

- Providing responses to the hook
- Using two small balls of clay or plasticine to represent two hydrogen atoms
- Inserting one tooth pick between them to represent one pair of shared electrons.
- Using a paper, pen and glue stick, label the toothpick as a single covalent bond
- Using two medium balls of clay or plasticine to represent two oxygen atoms
- Inserting two tooth picks between them to represent two pairs of shared electrons
- Labelling it as a double covalent bond
- Using two medium balls of clay or plasticine to represent two nitrogen atoms
- Inserting three tooth picks between them to represent three pairs of shared electrons
- Labeling it as a triple covalent bond
- Explaining in terms of electrons how the bonds in the models of hydrogen, oxygen and nitrogen are formed
- Drawing the electron diagrams showing the bonding in the models they constructed
- Making presentations
- Displaying their models in the classroom

Assessment Guide:

- Check if learners are:
    - constructing the models correctly
    - describing the bonding in the molecules correctly
    - drawing the electronic diagrams correctly

Key Learning Points:

Atoms bond to attain a stable electron configuration like noble gases. Non-metals such as oxygen and nitrogen share electrons because they have high electronegativity and cannot easily loose or gain electrons. The number of pairs of electrons shared between atoms determine the nature of the covalent bond formed.

- Single covalent bond: Formed by one pair of shared electrons. For example, hydrogen (H_2)

  H – H
- Double covalent bond: Formed by two pairs of shared electrons. For example, oxygen, (O_2)

  O = O

- Triple covalent bond: Formed by three pairs of shared electrons. For example, nitrogen (N_2)
  $N \equiv N$

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating

Activity 3: Explaining the properties of covalent compounds

In this activity, learners will work in small groups to explain the properties of covalent compounds.

HOOK: Imagine that you are provided with the following substances: ethanol, paraffin, salt solution, lemon juice and diesel. You then use them to complete an electrical circuit connected to a light bulb. Explain what you would observe with each of the substances.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Flip charts, markers, bostik, worksheet showing the properties of some selected covalent compounds

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with a worksheet
- Ask learners to:
  - explain the trend in melting and boiling points of covalent compounds
  - explain why covalent compounds do not conduct electricity in both solid and solution form
  - explain the lower density of covalent compounds
  - discuss and share findings with classmates

Learners' Activities:

- Providing responses to the hook.
- Completing the following worksheet by writing does not conduct or conducts

Compound	Melting Point/^{0}C	Boiling Point/^{0}C	Density g/cm^3	Electrical Conductivity
Carbon dioxide	-78	-78	-0.00198	
Methane	-182	-161	-0.000656	
Ethanol	-144	78	0.789	
Paraffin	-24	301	0.83	

- Explaining:
    - the trend in melting and boiling points of covalent compounds.
    - why covalent compounds do not conduct electricity in solid and solution form
    - why covalent compounds have lower density
- Discussing and making presentations to the class

Assessment Guide:

- Check if learners are explaining the properties of covalent compounds correctly

Key Learning Points:

The following are some of the properties of covalent compounds:

- They are made up of molecules. Covalent compounds exist as discrete molecules rather than extended lattices.
- They have low melting and boiling points because of the weak forces of attraction which hold the molecules. Little energy is needed to break them apart.
- They are insoluble in water but soluble in organic solvents such as ethanol and petrol.
- They do not conduct electricity in solid or solution form because they are made up of molecules. There are no free electrons or ions to carry charge in any state
- They have low densities. Molecules in covalent compounds are spaced further apart compared to tightly packed ions in ionic compounds.
- They are often gases or liquids at room temperature and pressure, but some are soft solids.

Skills Developed:

- Analytical Thinking
- Collaborating

- Communicating

Activity 4: Describing a molecule
In this activity, learners will work in small groups to describe a molecule.

HOOK: Consider the wall as a chemical substance.

Can the bricks exist on their own even without the wall?
What would the bricks represent in the chemical substance?

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Computers connected to internet, flip charts, markers, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put the learners in small groups
- Distribute the suggested or any improvised materials
- Ask learners to:
    - use digital devices, science dictionary or textbooks to search for the description of a molecule
    - differentiate a molecule from a compound
    - discuss findings with classmates for peer review
    - write findings in exercise books
    - make presentations to the class

Learners' Activities:

- Providing responses to the hook
- Searching for the description of a molecule using digital devices, science dictionary or textbooks
- Differentiating a molecule from a compound

- Discussing findings with classmates for peer review
- Writing findings in exercise books
- Making presentations to the class

Assessment Guide:

- Observe learners' participation during the activity
- Check if learners are:
    - describing a molecule correctly
    - differentiating a molecule from a compound correctly

Key Learning Points:

- A molecule is a group of covalently bonded atoms which can exist independently. A molecule can also be described as a smallest particle of an element or compound that exists independently.
- Molecules consist of non-metal atoms held by covalent bonds. **There are two types of molecules; molecules of elements and compound molecules. Molecules** of elements consist of two or more atoms of the same element (e.g. O_2, O_3 and N_2). **Compound molecules** are made up of two or more different elements (e.g. H_2O and CO_2).

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating

Activity 5: Exploring the uses of covalent compounds

In this activity, learners will work in small groups to explore the uses of covalent compounds.

HOOK: Think about the water you drink, the sugar that sweetens your tea, and oxygen you breath. These are all covalent compounds. Think of other covalent compounds and what they are used for.

Suggested Teaching and Learning Materials:

- **Natural Materials**: Water, sugar, silicon dioxide, ethanol, petrol, diesel, fats and oils
- **Artificial Materials**: Aspirin, plastics (polymers e.g. PVC), rubbers

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the lesson using the hook
- Put learners in small groups and provide the materials
- Ask learners to:
    - identify the nature of materials provided in relation to bonding
    - discuss the uses of the materials provided
    - share more examples of covalent compounds used in everyday life
    - present findings
- Consolidate presentations from learners

Learners' Activities:

- Providing responses to the hook
- Identifying the nature of materials provided in relation to bonding
- Discussing the uses of the materials provided
- Sharing more examples of covalent compounds used in everyday life
- Presenting findings

Assessment Guide:

- Observe learner participation during the activity
- Assess learners with questions on the uses of different covalent compounds

Key Learning Points:

Covalent compounds have a wide range of uses. The following are some of the covalent compounds and their uses:

- Carbon dioxide is used in fire extinguishers as well as in carbonated drinks
- Ethene is used in making polythene plastics
- Methane is used for heating in gas stoves
- Petrol and diesel are used as fuels to run engines
- Ethanol is used as an alcoholic beverage, a liquid in thermometers and as an organic solvent.
- Water is a universal solvent. It is used as a solvent to dissolve various ionic compounds.

Skills Developed:

- Analytical Thinking
- Collaborating
- Communicating
- Observing

- Problem solving

Expected Standards:

- Understanding of covalent bonding demonstrated correctly.
- The uses of covalent compounds applied in everyday life accordingly.

Sub-Topic 2.1.6: Metallic Bonding

Introduction:

Metallic bonding is a type of chemical bonding that occurs in metals, where positively charged metal ions are surrounded by a "sea" of delocalised electrons. These free-moving electrons hold the metal atoms together, giving rise to unique properties such as high electrical conductivity, malleability, ductility, and lustre. Understanding metallic bonding helps explain why metals are strong yet flexible, and why they play a vital role in everyday life like in construction and technology.

Specific Competences:

2.1.6.1 Demonstrate understanding of metallic bonding

2.1.6.2 Apply the uses of metallic substances in our everyday life

Key Terms:

- **Delocalised electrons**: Free electrons within the metallic lattice
- **Electrostatic force**: Force of attraction between oppositely charged particles
- **Metallic lattice**: A regular arrangement of positively charged ions surrounded by a sea of electrons

Learning Activities:

There are four activities on this sub-topic to help learners demonstrate understanding of metallic bonding.

Activity 1.0: **Describing metallic bonding**

This activity will involve learners researching on what metallic bonding is by searching from various materials in small groups.

HOOK: Ionic bonding → metal + non-metal

Covalent bonding → non-metal + non-metal

Metallic bonding → ???

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Computers connected to internet, textbooks containing information on different types of bonding

Learning Environment Setup:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Ask learners to:
  - find out as much information as possible from the sources available about metallic bonding
  - make presentations
- Consolidate presentations from learners

Learners' Activities:

- Finding out as much information as possible from the sources available about metallic bonding
- Making presentations
- Adding anything left out and asking questions for clarity

Assessment Guide:

- Check correctness of information being presented by learners

Key Learning Points

- Metals are electropositive elements
- Metallic bonding involves delocalized valence electrons and positively charged ions in a metallic lattice
- The delocalized electrons hold the structure together

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 2.0: Exploring the uses of metallic substances

There are two activities that will help learners to explore the uses of metallic substances.

Activity 2.1: Exploring the uses of metallic substances by research

In this activity, learners will engage in research to explore the uses of metallic substances.

HOOK: Analyse the following objects:

Roofing sheets, pots, heating elements, electrical cables

If you were tasked to recommend the suitable materials to be used to make each of the objects, what would be your recommendation?

Suggested Teaching and Learning Materials:

- **Natural Materials**: Iron, copper, aluminium, gold, silver
- **Artificial Materials**: Flip charts, bostik, textbooks, computers connected to internet, markers

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity by showing everyday metallic objects (e.g., coins, aluminium cans, copper wires, steel utensils)
- Divide learners into small groups
- Assign each group a specific metal (e.g., iron, copper, aluminium, gold, silver)
- Provide resources (charts, textbooks, internet access, or handouts) for research

Learners' Activities:

- Researching the uses of the assigned metals in different fields (construction, medicine, electronics, jewellery, transportation)
- Working together to prepare a short presentation showing how the metal is used in real life
- Sharing findings with the class

Assessment Guide:

- Assess group presentations based on accuracy and ability give practical applications

Key Learning Points:

Metallic substances have several individual uses. The following are the uses of some metallic substances:

- Iron-roofing sheets, construction beams, car bodies
- Steel-construction beams, tools
- Aluminium -Aircraft bodies, packaging, utensils
- Copper-wiring, plumbing, electronics

- Gold-Jewellery, electronics

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 2.2: Investigating the properties of metals

In this activity learners will investigate the physical properties of metallic substances through simple experiments.

HOOK: You are provided with a number of items made of different materials such as plastics, wood, metals, and glass. Discuss the various ways in which you can identify metallic ones.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Iron nails, copper wire, aluminium foil, steel spoon, a battery, bulb, wires, magnets, and a hammer

Learning Environment Set-up:

- **Natural Environment:** School surroundings

Artificial Environment: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide the materials to learners
- Ask learners to test each metal sample for:
  - Conductivity: Connect metal into a circuit with a bulb.
  - Malleability: Tap gently with a hammer to see if it flattens or bends.
  - Magnetism: Bring a magnet close to check attraction.
- Ask learners to:
  - Note down observations in a table (e.g., "Copper – good conductor, malleable, non-magnetic").
  - Share findings with the class

Caution*: Ensure all experiments are safe (no sharp edges, proper handling of wires, controlled use of electricity)*

Learners' Activities:

- Testing each metal sample for:
  - Conductivity: Connect metal into a circuit with a bulb.
  - Malleability: Tap gently with a hammer to see if it flattens or bends
  - Magnetism: Bring a magnet close to check attraction.
- Noting down observations in a table (e.g. copper - good conductor, malleable, non-magnetic)
- Sharing findings with the class

Assessment Guide:

- Check if learners are:
  - Conducting the tasks safely and correctly
  - Making correct observations

Key Learning Points:

Some of key properties of metals include the following:

- High electrical conductivity due to easy flow of electrons
- Thermal conductivity due to vibration of particles and the free movement of electrons
- Malleability and ductility: Layers of atoms slide past each other, enabling shaping into sheets or wires.
- High melting points and density: Strong electrostatic forces of attraction that require more to break

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 3: Justifying the uses of specific metallic substances for different applications in everyday life

This activity challenges learners to analyse why specific metals are chosen for particular uses. For example, copper for electrical wiring, aluminium for aircraft bodies, iron or steel for construction, and gold for jewellery. In this activity learners will research, justify selections, and debate alternatives while linking metal properties to practical applications.

HOOK: At Mopani copper mines, an electrical engineer making a distribution board must choose a metal conductor that carries electricity efficiently and effectively. The Engineer compares metals like iron, aluminium, brass, copper and cobalt. Using factors such as electrical conductivity, heat generation, strength, flexibility and resistance to corrosion. After evaluating these properties, copper is chosen as ideal for wiring the distribution boards.
Brainstorm why copper is preferred.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Printed images of objects such as wires, cans, bridges, jewellery, textbooks or simple online references, flip charts and markers, bostik

Learning Environment Setup:

- **Natural Environment:** school surroundings
- **Artificial Environment**: Classroom/Computer Laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Ask learners to:
    - select 4 assigned metals (copper, aluminium, iron/steel, gold) and give 2 everyday applications for each
    - List key properties justifying the choice
    - Create and complete a table on the flip chart like the one below:

S/N	**Metal**	**Use/Application**	**Justification**
1.	Copper	Electrical wiring	Conducts electricity, bends without breaking
2.			
3.			
4.			

-
    - Present the charts to the class
- Initiate a class debate on justifications in terms of environmental sustainability, cost implications and other factors

- Wrap up with key takeaways on property-application matches

Learner's Activities:

- Selecting 4 assigned metals (copper, aluminium, iron/steel, gold) and give 2 everyday applications for each
- Listing key properties and justifying the choice
- Creating and completing a table on the flip chart as guided by the teacher
- Presenting the charts to the class
- Debating on justifications in terms of environmental sustainability, cost implications and other factors

Assessment Guide:

- Assess using a rubric (justification clarity 40%, property links 30%, creativity 20%, participation 10%)

Extension:

- Teacher can organise a Field trip to visit a local workshop such as a welding shop to observe metal uses

Key Learning Points:

The table below shows primary uses of some metallic substances and justification for each use:

Metal/ Alloy	Primary uses	Justification
Iron/ Steel	Construction, tools, vehicles	Strength and durability from robust bonding withstand loads.
Aluminium	Aircraft, packaging, utensils	Lightweight yet strong; corrosion-resistant for longevity, malleability
Copper	Wiring, plumbing, electronics, ornaments	Excellent conductivity from delocalized electrons, resistance to corrosion, lustre, ductility
Gold	Jewellery, electronics	Ductility and non-reactivity which prevent tarnishing, lustre, malleability

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standards:

- Conceptual understanding of metallic bonding demonstrated correctly
- The uses of metallic substances in our everyday life applied correctly

Sub-Topic 2.1.7: Structure of Compounds

Introduction:

The structure of compounds refers to the spatial arrangement and bonding of atoms within molecules. Compounds form through chemical bonds such as ionic and covalent, creating unique three-dimensional structures. These structures determine properties such as melting points, solubility, and reactivity. Mastering compound structures enables learners to predict reactions, understand real-world applications like drug design, and interpret diagrams in exams. It builds critical thinking for advanced topics like organic chemistry and materials science, essential for scientific careers.

Specific Competence:

2.1.7.1 Classify compounds into simple and giant structures.

Key Term:

- **Macromolecule**: These are large/giant molecules formed from many smaller units

Learning Activities:

Activity 1: **Describing structures of simple and giant compounds**

This hands-on activity helps learners explore and describe structures of simple molecular compounds such as water and methane in relation to giant compounds such as diamond, graphite and sodium chloride by building models. They will further discuss their properties. It builds visual understanding of bonding types and reinforces why structures determines properties like melting points.

HOOK: Have you ever wondered why water flows easily while diamond is one of the hardest substances on earth? Both are made from atoms yet the way the atoms are arranged is different.
Brainstorm on how atoms are arranged in the two substances.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Marshmallows/playdough/clay, toothpicks, coloured markers, worksheets, projector for introduction visuals

Learning Environment Setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Ask learners to:
  - research on structures of simple and giant molecules
  - draw and label models of simple molecules and ionic lattices e.g.H_2O V-shaped, CO_2 linear, NH_3 pyramidal
  - draw and label models of giant molecules and ionic lattices (e.g. diamond, sodium chloride, silicon dioxide, graphite) noting bond types and shape
  - discuss and predict properties
  - present the work

Learner's Activities:

- Researching on structures of simple and giant molecules
- Drawing and label models of simple molecules and ionic lattices e.g.H_2O V-shaped, CO_2 linear, NH_3 pyramidal
- Drawing and labelling models of giant molecules and ionic lattices (e.g. diamond, sodium chloride, silicon dioxide, graphite) noting bond types and shape
- Discussing and predicting properties
- Presenting the work

Assessment Guide:

- Use a checklist (structure accuracy, property links) and note misconceptions if any

Key Learning Points:

Simple molecules

- Simple molecules are discrete molecules held by weak intermolecular forces e.g., H_2O, CO_2 and NH_3
- Generally, they have low melting and boiling points, and are liquids or gases at room temperature and pressure

Giant structures

- Macromolecules: Large molecules that are covalently bonded e.g., diamond, graphite, silicon dioxide. They have strong covalent bonds between atoms. They have high melting and boiling points, are usually non-conductors except graphite, and are usually solids at room temperature and pressure.

- Ionic lattices: These are large ionic structures e.g., NaCl, MgO and CaF_2. They have strong electrostatic forces of attraction between ions. They are solids at room temperature and pressure. They have high melting and boiling points, are hard and brittle and conduct electricity when in molten or solution form.

Skills Developed:
- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 2: **Illustrating structures of Simple and Giant Compounds**
This activity involves learners building up models of simple and giant structures.

HOOK: Imagine the gas you breath out, CO_2, floating invisibly in the air. And think of diamond, one of the hardest substances on earth, and table salt (NaCl) which forms tiny crystals in the kitchen. These substances look and behave differently yet they are all made from atoms joined in specific structures.
You are going to explore how atoms arrange themselves into simple structures (e.g. CO_2) and giant structures (e.g. diamond and NaCl)

Suggested Teaching and Learning Materials:
- **Natural Materials**:
- **Artificial Materials**: Coloured modelling clay or marshmallows (for atoms), toothpicks or pipe cleaners (for bonds), flip chart, markers, and rulers, printed reference cards with compound examples (e.g., H_2O, NaCl, SiO_2, graphite), whiteboard and projector for class sharing

Learning Environment Setup:
- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:
- Introduce the activity
- Put learners in small groups
- Provide learners with the materials

- Ask learners to build/construct:
    - simple molecules (e.g. H_2O bent, CO_2 linear, NH_3 pyramidal) using marshmallows/clay and toothpicks or atomic models
    - giant molecules (e.g. NaCl, MgO, CaF_2) using marshmallows/clay and toothpicks or atomic models
    - present to class

Learner's Activities:

- Building/constructing simple molecules (e.g. H_2O bent, CO_2 linear, NH_3 pyramidal) using marshmallows/clay and toothpicks or atomic models
- Building/constructing giant molecules (e.g. NaCl, MgO, CaF_2) using marshmallows/clay and toothpicks or atomic models
- Presenting to class

Caution: *Do not taste or eat the materials provided*

Assessment Guide:

- Check if learners are building structures correctly
- Assess informally through observations and plenary questions; collect models/ illustrations for marking
- Use an assessment rubric to assess the learners

Sample Assessment Rubric

Criterion	**Excellent (4)**	**Good (3)**	**Needs Work (1-2)**
Accuracy of structures/bonds	Precise atoms, bonds, forces shown	Mostly correct	Major errors
Labelling and properties	Clear labels + 2 relevant properties	Basic labels	Incomplete
Explanation/Participation	Confident, links structure to properties	Basic explanation	Minimal input

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 3: **Distinguishing simple and giant structures by their models**

This activity uses physical models that were built in the previous activity to help learners visualize and compare simple and giant structures.

HOOK: Sugar easily melts on heating, and dissolves in water. Candle wax easily melts on heating but does not dissolve in water. Table salt dissolves in water but does not easily melt on heating.
Discuss and explain the differences amongst the three substances.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Marshmallows/mini marshmallows (atoms), toothpicks/spaghetti (bonds), coloured paper/markers for drawing, printed structure cards with images of I_2, CO_2, diamond, NaCl lattice), worksheets

Learning Environment Setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Give the groups models built in the previous activity
- Ask learners to use their models to distinguish:
    - simple and giant structures
    - macromolecules and ionic lattices
- Ask learners to present the work
- Conclude by leading the class in a plenary discussion on real-world links 'why diamonds are hard gems'

Learner's Activities:

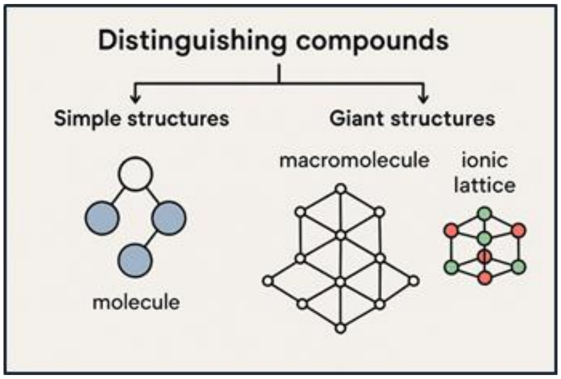
- Distinguishing simple and giant structures
- Distinguishing macromolecules and ionic lattices
- Presenting the work

Assessment Guide:

Review group presentations; take note of learner participation and accuracy of concepts

Key Learning Points:

The illustration below distinguishes structurally, simple structures and giant structures.



The Table below summarises key concepts of simple covalent molecules, giant covalent molecules and ionic lattices.

Structure Type	Bonding Type	Example	Key Properties
Simple molecules	Covalent and weak intermolecular forces	CO_2, H_2O	Low melting points and boiling points, non-electrical conductors
Giant molecules	Strong covalent network	Diamond, Graphite	Very high melting points and boiling points, hard/ soft, mostly non-conductors
Ionic Lattice	Electrostatic attraction	NaCl, MgO	high melting points and boiling points, brittle, conduct when molten/aqueous

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standard:

- Compounds classified into simple and giant structures accordingly

TERM 2

TOPIC 2.2: CHEMICAL REACTIONS

Introduction:

Chemical reactions are the heart of Chemistry. The essence of chemistry is to study chemical reactions. They involve the combination of the elements and their compounds to give new substances. Chemists have used chemical reactions to produce a range of various materials which are so vital in life. Plants, animals and even ourselves are chemical reactions factories. Chemical reactions are different in nature ranging from simple to complex ones. However, all chemical reactions are governed by some fundamental principles some of which will be discussed in this topic. Some of the principles include; types, representation and balancing of chemical reactions.

General Competences:

- Analytical Thinking
- Collaboration
- Communication
- Critical Thinking
- Environmental Sustainability
- Problem Solving

Sub-Topic 2.2.1: Chemical Reactions

Introduction

Elements and compounds interact in a variety of ways to form new substances. These interactions can be categorised using aspects such as, ways of combinations or disintegrations, and energy involvement. This subtopic will look at the various types of chemical reactions.

Specific Competence:

2.2.1.1 Demonstrate understanding of the principles of chemical reactions

Key Terms:

- **Chemical reaction:** A process where substances called reactants chemically transform into new substances called products.
- **Reactant**: A substance present at the start of a chemical reaction
- **Product**: This is a new substance formed after a chemical reaction has taken place

Learning Activities:

The following activities will help learners to describe a chemical reaction and to investigate the types of chemical reactions.

Activity 1.0: Describing a chemical reaction and investigating the types of chemical reactions

In this activity, learners will engage in a reteach to describe a chemical reaction and types of chemical reactions. They will further demonstrate the types of chemical reactions through a hands-on task.

HOOK: At Ndola lime industry in Ndola, lime is manufactured by strongly heating limestone.

- What happens when limestone is heated?
- What new substances are formed?
- In which other ways can substances react to produce new substances?

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Text books, science dictionary, paper letter cuttings (A, B, C, D, E, F, AB, CB, AD), flip charts, markers, bostik, computers connected to internet

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry or computer laboratory

Teacher's Roles:

- Introducing the lesson using the hook
- Put learners in small groups
- Provide the materials to the learners
- Ask learners to:
  - use a science dictionary and other reference materials like textbooks to search for the meaning of chemical reaction and to research on the types of chemical reactions
  - describe a chemical reaction and the different types of chemical reactions
  - use the paper letter cuttings to formulate hypothetical different chemical reactions on the flip chart e.g. $A + B \rightarrow AB$
  - present work to the class
- Consolidate on the presentations by the learners

Learners' Activities:

- Searching for the meaning of chemical reaction and to research on the types of chemical reactions using a science dictionary and other reference materials like textbooks and digital devices
- Describing a chemical reaction and the different types of chemical reactions
- Using the paper letter cuttings to formulate hypothetical chemical reactions

- Presenting work to the class

Assessment Guide:

- Check if learners are:
    - describing chemical reactions and the types correctly
    - formulating hypothetical reactions correctly

Key Learning Points:

- A chemical reaction is the process of bond breaking and bond forming. The bonds of the reactants break while new bonds form producing new substances called products. Chemical reactions are represented by chemical equations in which reactants are written on the left side of the arrow and the products on the right side of the arrow.
- The process of forming products from reactants can take place in a number of ways. These different ways give the types of chemical reactions as shown below:
    - Direct Combination or synthesis reaction – this is the type of reaction in which two or more elements/compounds combine to form one product as shown**:**

      $$\mathbf{A + B \rightarrow AB}$$
    - Decomposition reaction – this is the type of reaction in which one substance breaks up or splits up forming two or more products. Usually, heat is involved

      $$\mathbf{AB \rightarrow A + B}$$

      This is the opposite a synthesis reaction.
    - Single displacement – this is a reaction in which a more reactive element replaces a less reactive element from its compound as shown:

      $$\mathbf{A + BC \rightarrow AC + B}$$
    - Double displacement reaction – this is also known as double decomposition. This is a type of reaction where two compounds (usually ionic) in aqueous solution exchange ions or radicals to form new compounds. The compounds first decompose and then, exchange the radicals as shown:

      $$\mathbf{AB + CD \rightarrow AD + CB}$$
    - Chain reaction – this is a reaction in which the reactants produce reactive products (intermediate products) which in turn engage in further reactions

      $$\mathbf{AB \rightarrow A + B}$$
      $$\mathbf{B + E \rightarrow BE}$$
      $$\mathbf{BE + F \rightarrow FE}$$

      Some chain reactions terminate out, while others are non-terminating.

Skills Developed:

- Communicating
- Collaborating
- Analysing

Expected Standard:

- Understanding of the principles of chemical reactions demonstrated correctly

Sub-Topic 2.2.2: Chemical Equations

Introduction:

The conversions of reactants into products are referred to as chemical reactions. These chemical reactions are represented by chemical equations. Chemical equations are a shorthand method of showing what happens in a chemical reaction. This shorthand method uses chemical symbols and formulae of elements and compounds respectively.

Specific Competence:

2.2.2.1 Construct chemical equations

Key Terms:

- **Word equation:** This is a representation of a chemical reaction in words
- **Chemical equation**: This is a symbolic shorthand representation of a chemical reaction
- **State symbols**: These are symbols that are used to show the physical state of the substances in a chemical reaction (equation)
- **Ionic equation**: An equation only showing the species that underwent change during the reaction
- **Spectator ions:** These are ions that did not undergo change during a reaction
- **Exothermic reaction**: A reaction that releases energy to the environment
- **Endothermic reaction**: A reaction that absorbs energy from the environment

Learning Activities:

The following activities will help learners to describe a chemical equation and its notation.

Activity 1.0: Describing a chemical equation and its notation

In this activity, learners will describe a chemical equation and its notation by researching from various sources of information.

HOOK: During baking, when baking soda is mixed with vinegar, bubbles are seen.

- Which chemical substances are reacting, and what is being produced?
- How can the reaction be represented?

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Text books, science dictionary, flip charts, markers, bostik, digital devices connected to internet

Learning Environment Set-up:

- **Natural Environment:** School surroundings

- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introducing the lesson using the hook
- Put learners in small groups
- Ask learners to:
    - use a science dictionary, digital devices and other reference materials provided like textbooks to search for the meaning of a chemical equation
    - describe how a chemical equation is written using hypothetical symbols
    - explain why an equal sign is not used to separate the reactants from the products in a chemical equation
    - describe a reversible reaction and demonstrate how it is written
    - present work to the class
- Consolidate on the presentations by the learners

Learners' Activities:

- Searching for the meaning of a chemical equation using a science dictionary and other reference materials provided
- Describing how a chemical equation is written using hypothetical symbols
- Explaining why an equal sign is not used to separate the reactants from the products in a chemical equation
- Describing a reversible reaction and demonstrating how it is written
- Presenting work to the class

Assessment Guide:

- Check if learners are describing a chemical equation and its notation correctly

Key Learning Points:

- A Chemical equation is a representation of a chemical reaction. It uses chemical symbols and formulae of the reactants and the products to show what happens in a chemical reaction. It consists of the reactants on the left and the products on the right, separated by a full headed arrow. For example, if AB reacts with CD to produce AD and CB in a double displacement reaction, the chemical equation would be written as:

$$\mathbf{AB + CD \rightarrow AD + CB;}$$

  where AB and CD are the reactants while AD and CB are the products. This is read as AB reacts with CD producing AD and CB. Note that the equal sign is not used to separate the reactants and the products. This is because the reactants and the products are chemically not equal, and thus the arrow should not be read as 'equal to.'

- However, some chemical reactions are reversible. This means that they can go either way by changing conditions. For such reactions, double half-headed arrows are used to denote reversibility. For example, a reversible reaction can be written as shown:

$$\mathbf{AB \rightleftharpoons A + B}$$

Skills Developed:

- Communicating
- Collaborating
- Analysing

Activity 2.0: Constructing word equations from descriptive chemical changes

This activity will help learners to construct word equations of chemical reactions.

> **HOOK**: During laboratory experiment, students mix a shiny strip of magnesium with dilute hydrochloric acid. They watch it fizz vigorously as a colourless gas escapes and the metal vanishes.
>
> Think of a very brief statement you can use in words to explain the reaction that was taking place.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Text books, work charts with descriptive chemical changes, flip charts, markers, bostik

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introducing the lesson using the hook
- Put learners in small groups
- Ask learners to:
    - construct word equations using the descriptions of the chemical changes provided on the work charts

      The example of a work chart to be prepared for this lesson is as shown below. The teacher to choose which reactions and the number to use. The first one has been done as an example

S/N	Description of a Chemical Change	Word Equation
1	Hydrogen gas reacts with oxygen gas to form water	Hydrogen gas + Oxygen gas $\rightarrow$ Water
2		
3		
4		
5		
6		
7		
8		
9		
10		

  - present work to the class
- Consolidate on the presentations by the learners

Learners' Activities:

- Constructing word equations using the descriptions of the chemical changes provided on the work charts
- Presenting work to the class

Assessment Guide:

- Check if learners are constructing word equations correctly

Key Learning Points:

- A word equation uses names of the substances in a reaction. The arrow is used to separate the reactants and the products. For example, given the description that hydrogen gas reacts with oxygen gas to form water, the word equation for this reaction can be written as:

  Hydrogen gas + Oxygen gas $\rightarrow$ Water

Skills Developed:

- Communicating
- Collaborating
- Analysing

Activity 3.0: Formulating and balancing chemical equations

This activity will involve learners in research leading to formulation of balanced chemical equations, with state symbols correctly written.

HOOK: In a fireworks display, leaners ignite a mixture of charcoal, sulphur, and potassium nitrate, which produces a burst of hot gases and solid residue. They formulate a word equation as;
Potassium nitrate + charcoal + sulphur → potassium sulphide + carbon dioxide + nitrogen

This does not show the exact number of atoms of each substance involved. More information can be reviewed in chemical equations.

Let's explore chemical equations!

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Text books, work charts with word equations, flip charts, markers, bostik, computers connected to internet

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry or Computer laboratory

Teacher's Roles:

- Introducing the lesson using the hook
- Put learners in small groups
- Ask learners to research on:
    - how chemical equations are written and balanced
    - state symbols and how they are indicated in a chemical equation
    - why chemical equations are supposed to be balanced
    - formulation of chemical equations using the word equations on the work charts, indicating the state symbols

    The example of a work chart to be prepared for this lesson can be as shown below. The teacher to choose which reactions and the number to use. The first one has been done as an example.

S/N	Description of a Chemical Change	Word Equation
1	Hydrogen gas + Oxygen gas → Water	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
2		
3		
4		

5		

  - present work to the class
- Consolidate on the presentations by the learners

Learners' Activities:

- Researching on:
  - how chemical equations are written and balanced
  - state symbols and how they are indicated in a chemical equation
  - why chemical equations are supposed to be balanced
  - formulating chemical equations using the word equations on the chart, indicating the state symbols
- Presenting work to the class

Assessment Guide:

- Check if learners are:
  - formulating chemical equations correctly
  - balancing the chemical correctly
  - indicating the state symbols correctly

Key Learning Points:

- Word equations are not enough to adequately show what goes on in a chemical reaction. They don't account for the atoms of each element involved. Therefore, chemical reactions are adequately represented using chemical equations. This is written by changing the names of each substance in a word equation into chemical symbols or formulae of the substances.

  For example:

  Hydrogen gas + Oxygen gas → Water

  $$H_2 + O_2 \rightarrow H_2O$$

- The equation is supposed to be balanced, in order to obey the law of conservation of matter which states that atoms are neither created nor destroyed in a chemical reaction. This is done by placing whole numbers in front of symbols or formulae of the substances in the equation. Thus, the above equation is balanced as shown:

  $$2H_2 + O_2 \rightarrow 2H_2O$$

- Further, the physical state of each substance taking part in the reaction is shown. This is done by writing state symbols written in lower case, in brackets. This should be in the same line immediately after each species. The state symbols are 's' for solid, 'l' for liquid, ‘g’ for gas and ‘aq’ for aqueous solutions. Thus, the complete chemical equation for the above reaction is;

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Skills Developed:

- Communicating
- Collaborating
- Analysing

Activity 4.0: Constructing net ionic equations from balanced chemical equations

This activity will help learners to construct net ionic equations from balanced chemical equations.

HOOK: After learning how to write chemical equations, a learner observes an experiment in which silver nitrate solution is reacted with sodium chloride, forming a white precipitate of silver chloride while sodium nitrate remains dissolved. The learner writes the chemical equation for the reaction as follows:

$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

The learner wonders whether there was truly a reaction as reactants appeared to have just exchanged ions. On asking the teacher of Chemistry, the learner was advised to research about ionic equations to identify the species that reacted.

Find out if all the ions reacted.

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Text books, flip charts, markers, bostik, computers connected to internet

Learning Environment Set-up:

- **Natural Environment**: School surroundings
- **Artificial Environment**: Classroom/Computer laboratory

Teacher’s Roles:

- Introduce the activity
- Put learners in small groups
- Ask learners to:
    - research on the rules for writing ionic equations

- list the rules for writing ionic equations with examples
- present work to the class

- Consolidate on the presentations by the learners

Learners' Activities:

- Researching on the rules for writing ionic equations
- Listing the rules for writing ionic equations with examples
- Presenting work to the class

Assessment Guide:

- Check if learners are:
  - listing the rules correctly
  - interpreting and applying the rules correctly
  - constructing the ionic equations correctly

Key Learning Points:

- A chemical equation does not adequately show what happens in a reaction. For example, it does not show which substances changed and which ones did not. Some species (ions) in a chemical reaction do not undergo chemical change. These ions are referred to as spectator ions. The ionic equation is written using only the ions that underwent change. These are the ions which are responsible for the observable change in a chemical reaction.
- The rules for writing ionic equations are as follows:
  - Write down the balanced chemical equation for the reaction with state symbols
  - Split solutions of acids and ionic substances into ions
  - Do not split into ions, the substances that are in solid, liquid and gaseous states
  - Cancel out spectator ions
  - Write down the ionic equation using the remaining species, carrying forward their state symbols
- For example, the reaction of sodium hydroxide solution reacting with dilute hydrochloric acid to give sodium chloride solution and water, its ionic equation would be obtained as shown following the above rules:

  The balanced chemical equation for the reaction is:

  $$NaOH(aq) \ + \ HCl(aq) \ \rightarrow \ NaCl(aq) \ + \ H_2O(l)$$

  Splitting the solutions of acid and ionic substances yields:

  $$Na^+(aq) \ + \ OH^-(aq) \ + \ H^+(aq) \ + \ Cl^-(aq) \ \rightarrow \ Na^+(aq) \ + \ Cl^-(aq) \ + \ H_2O(l)$$

  Cancelling out the ions that did not undergo change (spectator ions) gives

  $$\cancel{Na^+}(aq) \ + \ OH^-(aq) \ + \ H^+(aq) \ + \ \cancel{Cl^-}(aq) \ \rightarrow \ \cancel{Na^+}(aq) + \cancel{Cl^-}(aq) \ + \ H_2O(l)$$

Writing the ionic equation with the remaining species we get:

OH^- (aq) + H^+(aq) $\rightarrow$ H_2O(l), as the net ionic equation

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Identifying

Activity 5.0: Investigating chemical reactions as Endothermic or Exothermic

The following activity will involve a laboratory based practical. It will help learners to investigate some chemical reactions in terms of their energy involvement.

HOOK: Bath salts (ammonium salts) are dissolved in water for foot soak. The mixture absorbs heat from its surroundings, causing the temperature of the water to drop noticeably often feeling icy cold to the touch. Mixing baking soda with vinegar releases heat, warming the container used noticeably.

Uncover the energy change secrets.

Suggested Teaching and Learning Materials:

- **Natural Materials:** Water, magnesium or aluminium metal
- **Artificial Materials**: Heat resistant beakers, thermometers, flip charts, markers, bostik, sodium hydroxide pellets, 1M sodium hydroxide solution, 1M hydrochloric acid solution, ammonium nitrate crystals, urea fertilizer, baking soda, vinegar, work sheet

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the lesson using the hook
- Put learners in small groups
- Provide the learners with the materials including the work sheet

Sample Work Sheet

Schedule of experiments to be investigated

S/N	Instructions	Temperature (^{0}C)/ Observation	Conclusion
1	**a)** Put about 60ml water in a beaker, measure and record its temperature		
	b) Carefully add 2-3 spatula full sodium hydroxide pellets in the water while stirring, and feel the beaker using your hands		
	c) Record the stable temperature reached		
2	**a)** Put about 60ml water in a beaker, measure and record its temperature		
	b) Carefully add 2-3 spatula full ammonium nitrate crystals in water while stirring, and feel the beaker with your hands		
	c) Record the stable temperature reached		
3	**a)** Add about 60ml water in a beaker, measure and record its temperature		
	b) Carefully add 2-3 spatula full urea fertilizer in water while stirring, and feel the beaker with your hands		
4	**c)** Record the stable temperature reached		
	a) Put 30ml of 1M hydrochloric acid in a beaker, measure and record its temperature		
	b) Carefully and slowly pour 30ml of 1M sodium hydroxide solution into the acid while stirring, and feel the beaker with your hands		
	c) Record the stable temperature reached		
5	**a)** Put 40ml vinegar in a beaker and measure its temperature		
	b) Carefully pour 2-3 spatula full baking soda into the vinegar while stirring and feel the beaker with your hands		
	c) Measure and record the stable temperature reached		
6	**a)** Put 40ml of 1M hydrochloric acid in a beaker and measure its temperature		
	b) Carefully put 3cm of magnesium ribbon (or similar quantity of aluminium, iron fillings or zinc) into the acid while stirring and feel the beaker with your hands		
	c) Measure and record the stable temperature		

S/N	Instructions	Temperature (^{0}C)/ Observation	Conclusion
	reached		

- Ask learners to:
    - carry out the experiments/reactions as outlined on the work sheet with optimum care
    - record the temperature readings/observations
    - make conclusions on which reactions are exothermic or endothermic
    - present work to the class
- Consolidate on the presentations by the learners

Learners' Activities:

- Carrying out the experiments/reactions with optimum care
- Recording temperature readings/observations
- Making conclusions on which reactions are exothermic or endothermic
- Presenting work to the class

Caution*: Safety precautions should be observed when conducting the reactions*

Assessment Guide:

- Check if learners are:
    - practicing safety rule correctly
    - carrying out the experiments correctly
    - recording the observations correctly
    - making correct conclusions

Key Learning Points:

- When chemical reactions take place, energy is either liberated or absorbed.
- Reactions that release energy to the environment are called exothermic reactions, while those that absorb energy from the environment are referred to as endothermic reactions.
- Natural examples of exothermic reactions include radioactivity, rusting, respiration and combustion. An example of a natural endothermic reaction is photosynthesis.

Skill Developed:

- Communicating
- Collaborating
- Analysing
- Observing

Expected Standard:

- Chemical equations constructed accordingly

TOPIC 2.3: CHEMICAL KINETICS

Introduction:

Chemical kinetics is the branch of Chemistry that studies, among other things, the rates at which chemical reactions occur and the factors that influence these rates. It provides understanding of the rate at which reactants are converted into products. Chemical kinetics is vital in a number of fields, such as manufacturing, environmental conservation, mining, agriculture and in the development of more efficient energy sources. Controlling of rates of chemical reactions is essential for efficiency, safety, and sustainability of industrial processes.

General Competences:

- Analytical Thinking
- Collaboration
- Communication
- Critical Thinking
- Environmental Sustainability
- Problem Solving

Sub-Topic 2.3.1: Rates of Chemical Reactions

Introduction:

Rate of a chemical reaction is simply the speed at which a given reaction takes place. There are many factors which affect the rates of chemical reactions.

Specific Competences:

2.3.1.1 Demonstrate understanding of rates of chemical reactions

2.3.1.2 Investigate factors that affect the rates of chemical reactions

Key Terms:

- **Rate of chemical reaction:** The speed at which a given reaction takes place

Learning Activities:

There are six learning activities designed to help learners demonstrate understanding of rates of chemical reactions and investigate factors that affect the rates of chemical reactions.

Activity 1.0: Describing the rate of a chemical reaction

In this activity, learners will work in small groups to describe the rate of a chemical reaction.

HOOK: You are baking cookies for your school's science fair. You mix baking powder into the dough and place your cookies in the oven. Your cookies puff up in just 10 minutes. When your friend uses the same amount of baking powder in her batch, her cookies barely rise, taking over 30 minutes to show any change.

What's going on?

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials:** Text books, science dictionary, digital devices connected to internet

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Computer laboratory

Teacher's Roles:

- Introduce the lesson using the hook
- Put learners in small groups
- Ask learners to:
    - use a science dictionary and other reference materials like textbooks to search for the meaning of rate of chemical reaction
    - write the description of rate of chemical reaction in their exercise books
    - share findings with their peers for peer learning
- Consolidate presentations by learners

Learner's Activities:

- Responding to the hook
- Searching for the meaning of rate of chemical reaction using a science dictionary and other reference materials like textbooks
- Describing rate of chemical reaction from the findings
- Writing the description of rate of chemical reaction in exercise books
- Sharing findings with peers for peer learning

Assessment Guide:

- Check if learners are describing rate of chemical reaction correctly

Key Learning Points:

- Rate of a chemical reaction is the speed of a chemical reaction. It can be described as the amount of reactants used up or products formed per unit time.

  Rate of reaction = $\frac{\text{change in amount of reactants/products}}{\text{change in time}}$

- Rates of reactions are always positive. The formula used to calculate the rate of reaction will give a negative answer if the concentration of the substance is decreasing with time. When this is the case, the formula includes a negative sign.

Skills Developed:

- Communicating
- Collaborating
- Analysing

Activity 2.0: Experimenting on the rates of chemical reactions

The following activity is designed to help learners demonstrate the rate of chemical reactions.

HOOK: On a farm, a compost heap (plant matter acted on by microbes in cool, moist soil) breaks down slowly over months, turning waste into rich organic fertilizer. But at home, if you crush an egg shell and splash hot vinegar on it, fizzing CO_2 evolves fast, dissolving it in seconds. Some reactions proceed faster than others.
Let's experiment…

Suggested Teaching and Learning Materials:

- **Natural Materials:** Water
- **Artificial Materials:** 250ml beaker, 3cm piece of magnesium ribbon, 20cm^3 of 1.0M hydrochloric acid, stopwatch

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with 250ml beaker, 3cm piece of magnesium ribbon, 20cm^3 of 1.0M hydrochloric acid, stopwatch

- Use a work sheet to instruct leaners to:
    - measure 20cm^3 of 1.0M hydrochloric acid into a 250cm^3 beaker
    - place 3cm of magnesium ribbon into the acid and immediately start the stop watch
    - observe the bubbling and stop the watch when the magnesium ribbon completely reacts or when fizzing stops
    - record the time taken for the reaction to finish
    - discuss and share observations with their classmates for peer review
    - present findings to the class
- Ask learners questions such as the following as they conduct the experiment:
    - What did you observe happening to magnesium ribbon during the reaction?
    - How can the time taken for the magnesium to disappear help you describe the rate of chemical reaction?

Learners' Activities:

- Responding to the hook
- Collecting the required materials
- Measuring 20cm^3 of 1.0M HCl into a 250cm^3 beaker
- Placing 3cm of magnesium ribbon into the acid and immediately starting the stop watch
- Observing the bubbling and stopping the watch when magnesium completely reacts or when fizzing stops
- Recording the time taken for the reaction to finish
- Discussing and sharing observations with classmates for peer review
- Responding to questions

Assessment Guide:

- Check if learners are:
    - doing the experiment cautiously
    - making correct observations

Key Learning Points:

- The rate of chemical reaction is the speed at which reactants are used up or products are formed.
- When magnesium ribbon reacts with dilute hydrochloric acid, magnesium chloride and hydrogen gas are produced. The magnesium dissolves gradually. The time taken for magnesium to disappear shows how fast or slow the reaction is. A shorter reaction time means a faster rate while a longer reaction time means a longer reaction rate. The rate of

reaction can be observed by bubbling or disappearing of the solid. Measuring reaction time helps compare the rate of different reactions.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying

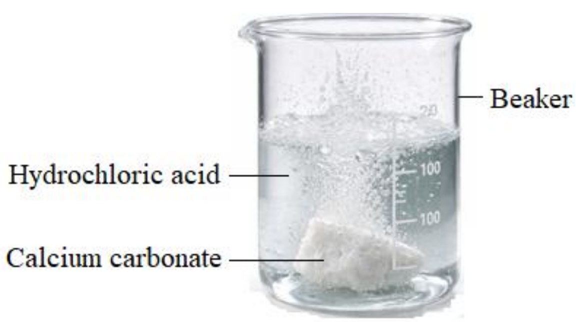
Activity 3.0: Applying rates of chemical reactions in real life situations

This activity will engage learners in comparing two reactions and relating them to real life situations.

> **HOOK**: A marketeer has green bananas and wants them to ripen fast for the following day's sale. The marketeer places green bananas in a closed space together with ripe apples (releasing ethylene gas). The bananas turn yellow in 2 days. The bananas that are left cool and alone are still green after a week.
> What does this tell you about the application of reaction rates?

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials:** A picture showing calcium carbonate reacting with an acid, and the process of ripening of bananas

 **A**	 **B**

Learning Environment Set-up:

- **Natural Environment**: School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with a picture showing two situations, A (calcium carbonate reacting with an acid) and B (process of ripening of bananas)
- Ask learners to:
    - discuss the chemical changes taking place in picture A and B
    - identify the process that happens quickly and the one that happens slowly
    - explain where else they can apply rates of chemical reactions in real life situations
    - discuss and share work with their classmates for constructive feedback

Learner's Activities:

- Providing responses to the hook
- Discussing the chemical changes taking place in picture A and B
- Identifying the process that happens quickly and the one that happens slowly
- Providing responses on the application rates of chemical reactions in real life situations
- Discussing and sharing work with classmates for constructive feedback

Assessment Guide:

- Check if learners are correctly:
    - identifying the processes taking place in the pictures and how fast they occur
    - relating the processes to real life situations

Key Learning Points:

The application of rates of chemical reactions in real life situations include the following:

- Food is cut into small pieces to make it cook faster
- Chewing of medicine tablets such as magnesium trisilicate or using powdered medicine to quicken absorption
- Catalysts help reduce air pollution in car exhausts
- Temperature and light affect photosynthesis in crops
- Lowering temperature slows food spoilage
- Controlling reaction rates improves industrial production like in Haber process
- Adjusting oxygen or surface area affects fuel combustion

Skills Developed:

- Communicating
- Collaborating

- Analysing
- Classifying

Activity 4.0: Exploring the factors that affect the rates of chemical reaction
This activity is divided into six sub-activities designed to help learners to explore the factors that affect the rate of chemical reaction. The factors that will be investigated include the following: temperature, concentration, surface area, pressure, catalyst and light.

Activity 4.1: Effect of temperature on the rate of a chemical reaction
In this activity, learners will investigate the effects of temperature on the rate of a chemical reaction.

HOOK: Same amount of sugar is put in two cups, one containing cold water and another containing hot water of the same volume. In which cup will the sugar dissolve faster and why? Discuss.

Suggested Teaching and Learning Materials:
- **Natural Materials**: Egg shells powder
- **Artificial Materials:** 1M Hydrochloric acid at room temperature, warm 1M hydrochloric acid, beakers, stop watch

Learning Environment Set-up:
- **Natural Environment**: School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:
- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with the materials
- Use a work sheet to instruct leaners to:
    - pour 100ml of dilute hydrochloric acid at room temperature into a beaker.
    - place one spatula full of egg shells powder into the beaker.
    - observe and record the time taken using a stop watch for the reaction to stop.
    - repeat steps 1, 2 and 3 using warm dilute hydrochloric acid.
- Ask learners the following questions as they do the task:
    (a) In which beaker did the egg shells powder dissolve faster?
    (b) What does this tell you about how temperature affects the rate of a chemical reaction?
    (c) Discuss and share your observations with classmates and ask groups to present the findings to the class

Learners' Activities:

- Providing responses to the hook
- Pouring 100ml of dilute hydrochloric acid at room temperature into a beaker.
- Placing one spatula full of egg shells powder into the beaker.
- Observing and recording the time taken using a stop watch for the reaction to stop.
- Repeating steps 1, 2 and 3 using warm dilute hydrochloric acid.
- Responding to questions, discussing and sharing observations with classmates and presenting findings to the class

Caution: *Observe safety precautions when handling chemical substances*

Assessment Guide:

- Check if learners are:
  - observing safety rules
  - making correct observations and recording them correctly
  - making correct conclusions

Key Learning Points:

- Temperature is the measure of the average kinetic energy of the particles
- When temperature increases, the rate of reaction also increases. This is because the particles gain more kinetic energy and move faster and collide more frequently and effectively. On the other hand, when temperature reduces, the rate of reaction also reduces. This is because the particles lose kinetic energy and move slower and do not collide frequently and effectively.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Identifying

Activity 4.2: Effect of concentration on the rate of a chemical reaction

In this activity, learners will investigate the effects of concentration on the rate of a chemical reaction.

HOOK: Imagine two markets. One market is over crowded with a lot of people, while the other market has very few people.

In which market will people be more likely to bump into each other as they move around shopping? Discuss.

Suggested Teaching and Learning Materials

- **Natural Materials:**
- **Artificial Materials**: Dilute hydrochloric acid of different concentrations (0.5M, 1.0M, 1.5M, 2.0M), dropper, stop watch, beakers, zinc powder/calcium carbonate

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide the materials to the learners
- Use a work sheet to instruct leaners to:
    - put 100ml of 0.5M dilute hydrochloric acid in a 250ml beaker.
    - add 1g of zinc powder to the acid in the beaker and stir, immediately start the stop watch.
    - record the time taken for the reaction to complete.
    - repeat steps 1, 2 and 3 with the other solutions of hydrochloric acid provided.
    - analyse the results and make conclusions
- Ask selected groups to make presentations
- Consolidate presentations from learners

Extended Activity

- Ask learners to plot a graph using the results obtained during the experiment

Learners' Activities:

- Responding to the hook
- Conducting the experiment:
    - Putting 100ml of 0.5M dilute hydrochloric acid in a 250ml beaker
    - Adding 1g of zinc powder to the acid in the beaker and stir, immediately start the stop watch

- Recording the time taken for the reaction to complete.
- Repeating steps 1, 2 and 3 with the other solutions of hydrochloric acid provided.

- Analysing the results and making conclusions
- Making presentations to the class

Assessment Guide:

- Check if learners are:
  - conducting the experiment correctly
  - collaborating
  - recording the results correctly

Key Learning Points:

- Concentration is the amount of solute per unit volume of solution
- When concentration increases, the number of successful collisions between particles resulting in the increase in the rate of reaction
- At lower concentration, there are fewer successful collision hence the reaction is slower

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 4.3: Effect of surface area on the rate of a chemical reaction

In this activity, learners will investigate the effect of surface area on the rate of a chemical reaction.

HOOK: During a chilly forest camping trip, two scout teams race to build a cooking fire. One piles big logs while the other splits the logs into smaller pieces.

Which team will light up their fire and cook faster? Explain.

Suggested Teaching and Learning Materials:

- **Natural Materials:** Water
- **Artificial Materials**: Beakers, measuring cylinder, electronic balance, stop watch, 1.0mol/dm^3 of hydrochloric acid, powdered calcium carbonate, chips of calcium carbonate

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Use a work sheet to instruct leaners to:
    - measure 60cm^3 of dilute hydrochloric acid and put it the beaker.
    - weigh 2g of powdered calcium carbonate and add it to the acid. Immediately start the stop watch.
    - record the time taken for the reaction to complete.
    - repeat steps 1, 2 and 3 with 2g of calcium carbonate chips.
    - analyse the results and making conclusions
    - make presentations to the class

Learner's Activities:

- Responding to the hook
- Conducting the experiment:
    1. Measuring 60cm^3 of dilute hydrochloric acid and put it the beaker.
    2. Weighing 2g of powdered calcium carbonate and add it to the acid. Immediately start the stop watch.
    3. Recording the time taken for the reaction to complete.
    4. Repeating steps 1, 2 and 3 with 2g of calcium carbonate chips.
    5. Analysing the results and making conclusions
- Making presentations to the class

Assessment Guide:

- Check if learners are:
    - conducting the experiment correctly
    - collaborating
    - recording the results correctly

Key Learning Points:

- Small particle size increases surface area for effective collisions and in turn increases the rate of a chemical reaction. On the other hand, large particle size reduces surface area for effective collisions and in turn reduces the rate of a chemical reaction.
- The powdered calcium carbonate react very fast and the reaction came to an end quickly. The calcium carbonate chips (marble chips) react slowly and take much longer to finish.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 4.4: Effect of pressure on the rate of a chemical reaction

In this activity, learners will research in small groups to explore the effect of pressure on the rate of a chemical reaction.

HOOK: During the synthesis of ammonia which is used in the manufacture of fertilizers, chemists raise pressure in high-pressure industrial reactors to force nitrogen and hydrogen molecules close together, speeding up the reaction to produce tonnes of the product in hours instead of days.

Find out more!

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Digital devices connected to internet, text books

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide the materials to learners

- Ask leaners to:
    - research on how pressure affects the rate of a chemical reaction
    - suggest and justify experiments that can be used to verify the effect of pressure on the rate of a chemical reaction
    - present their findings

Learner's Activities:

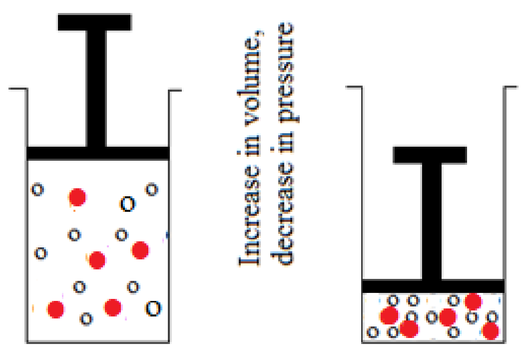
- Responding to the hook
- Researching on how pressure affects the rate of a chemical reaction
- Suggesting and justifying experiments that can be used to verify the effect of pressure on the rate of a chemical reaction
- Presenting findings to the class

Assessment Guide:

- Check for:
    - learner participation during the activity
    - correctness of information

Key Learning Points:

- Pressure affects chemical reactions involving gases
- When pressure increases, the rate of reaction also increases. This is because when pressure increases, the volume decreases making the gas particles to be closer and collide more frequently. On the other hand, when pressure reduces, the rate of reaction also reduces. This is because when pressure reduces, volume increases making the gas particles to be further apart and collide less frequently.



Low pressure, large volume   High pressure, small volume

Skills Developed:

- Analysing
- Classifying
- Collaborating
- Communicating
- Identifying

Activity 4.5: Effect of catalyst on the rate of chemical reaction

In this activity, learners will explore the effect of a catalyst on the rate of a chemical reaction.

HOOK: Imagine you are making fritters at home. You mix flour, water, sugar and yeast. The dough rises and there you go.
Brainstorm on the following:
a) What does yeast do?
b) What would happen if yeast is not added?

Suggested Teaching and Learning Materials:

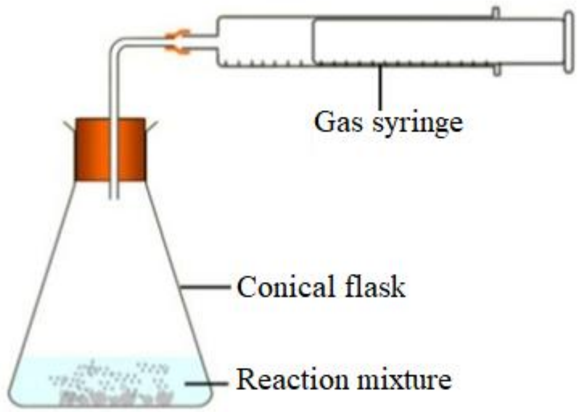
- **Natural Materials:**
- **Artificial Materials**: Hydrogen peroxide, manganese dioxide, beakers, gas syringe, delivery tube, stoppers, clamp stands, stop watch, conical flask

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide materials to learners
- Guide learners to arrange the apparatus as shown below:



- Use a work sheet to instruct leaners to:
    - put 100ml hydrogen peroxide in the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
    - record the volume of the gas produced every after 1 minute until effervescence stops
    - put a spatula full of manganese dioxide in another clean round-bottomed flask
    - add 100ml hydrogen peroxide to the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
    - record the volume of the gas produced every after 1 minute until effervescence stops
    - analyse the results and make conclusions
    - make presentations

Learner's Activities:

- Responding to the hook
- Putting 100ml hydrogen peroxide in the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
- Recording the volume of the gas produced every after 1 minute until effervescence stops
- Putting manganese dioxide in another clean conical flask
- Adding 100ml hydrogen peroxide to the round-bottomed flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
- Recording the volume of the gas produced every after 1 minute until effervescence stops
- Analysing the results and making conclusions
- Making presentations

Assessment Guide:

- Check:
    - learner participation during the activity

- correct handling of apparatus and chemicals
- for correct analysis and conclusions

Key Learning Points:

- A catalyst is a chemical substance which increases the rate of reaction but remains chemically unchanged at the end of the reaction
- A catalyst speeds up a chemical reaction by lowering the activation energy of the reaction. It lowers the amount of energy needed for successful collisions. Therefore, more collisions will be successful per unit time

Skills Developed:

- Analysing
- Classifying
- Collaborating
- Communicating
- Identifying

Activity 4.6: Effect of light on the rate of a chemical reaction

In this activity, learners will engage in a practical experiment to explore the effect of light on the rate of a chemical reaction.

HOOK: A laundry attendant washes clothes and dries them on a line. The line passes under the shade of a tree on one end. Some of the clothes are exposed to direct sunlight while others are in the shade.

Which clothes will dry faster? Discuss

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: 20 volume (6%) hydrogen peroxide solution, manganese dioxide, beakers, gas syringe, delivery tube, stoppers, clamp stands, stop watch, conical flask

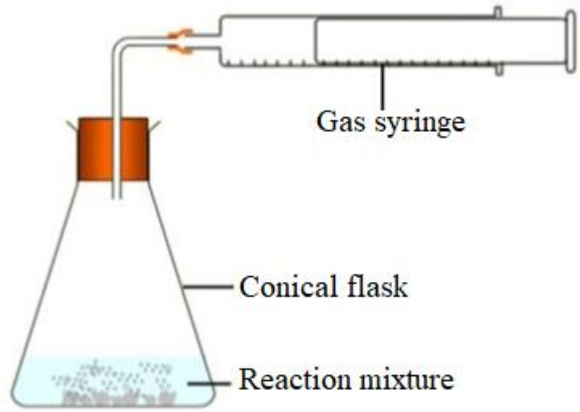
Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity

- Put learners in small groups
- Provide materials to learners
- Guide learners to arrange the apparatus as shown below:



- Use a work sheet to instruct leaners to:
    - put a spatula full of manganese dioxide in another clean conical flask
    - put 100ml hydrogen peroxide in the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
    - record the volume of the gas produced every after 1 minute until effervescence stops
    - wrap a clean round-bottomed flask with aluminium foil to block light. Add 100ml hydrogen peroxide to the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
    - record the volume of the gas produced every after 1 minute until effervescence stops
    - analyse the results and make conclusions
    - make presentations

 **Learner's Activities:**

- Responding to the hook
- Putting a spatula full of manganese dioxide in another clean conical flask
- Putting 100ml hydrogen peroxide in the conical flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
- Recording the volume of the gas produced every after 1 minute until effervescence stops
- Wrapping a clean conical flask with aluminium foil to block light. Add 100ml hydrogen peroxide to the round-bottomed flask, immediately connect the delivery tube leading to the gas syringe and start the stop watch
- Recording the volume of the gas produced every after 1 minute until effervescence stops
- Analysing the results and make conclusions
- Making presentations

Assessment Guide:

- Check:
    - learner participation during the activity
    - correct handling of apparatus and chemicals
    - for correct analysis and conclusions

Key Learning Points:

- The rate of a chemical reaction increases in the presence of light
- Light energy provides activation energy required for the reaction. Such reactions are referred to as photochemical reactions.
- An example a real-life photochemical process is photosynthesis, where plants use light energy to drive the reaction between carbon dioxide and water to produce glucose and oxygen

Skills Developed:

- Analysing
- Classifying
- Collaborating
- Communicating
- Identifying

Activity 5: Collecting and interpreting data on the rates of chemical reactions

In this activity, learners will engage in a practical experiment to explore the effect of concentration on the rate of a reaction between magnesium and hydrochloric acid. They are going to collect and interpret data on the rates of chemical reactions by graphical representations.

HOOK: Collecting and interpreting data on reaction rates reveals information in dynamic processes, guiding us to control them in laboratories and industry.

Precise data collection, using tools like timers and sensors, captures changes in concentration, mass, or volume over time.

Now, explore data collection and interpretation.

Suggested Teaching and Learning Materials:

- **Natural Materials:** Water
- **Artificial Materials**: Beakers, measuring cylinders, magnesium ribbon, hydrochloric acid solutions of different concentrations (0.5M, 1.0M, 1.5M, 2.0M), stop watch, graph paper

Learning Environment Set-up:

- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity using the hook
- Put learners in small groups
- Provide each group with four beakers, one measuring cylinder, 4 pieces of 3cm magnesium ribbon, hydrochloric acid solutions (0.5M, 1.0M, 1.5M, 2.0M), stop watch
- Use a work sheet to instruct leaners to:
    1. pour 20ml of 0.5M hydrochloric acid into a beaker.
    2. add a 3cm magnesium ribbon into the acid and start the stop watch immediately
    3. record the time taken for magnesium to completely react
    4. repeat steps 1, 2 and 3 for all other concentrations of dilute hydrochloric acid
    5. design a table for the presentation of your results

    Sample of the table of results

Concentration of HCl (M)	Volume of HCl (ml)	Length of magnesium ribbon (cm)	Time taken for magnesium to react (min)	Rate of reaction (cm/min)

    6. plot a graph of rate of reaction against concentration of HCl
    7. analyse and interpret the graph, and make conclusions
    8. make presentations to the class

Learners' Activities:

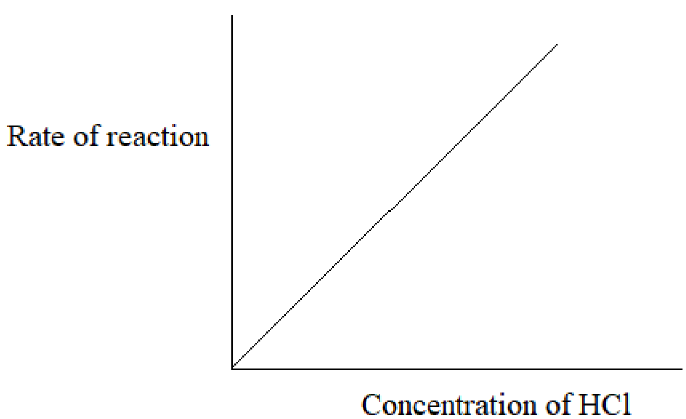
- Responding to the hook
- Conducting an experiment following the procedure given:
    - Pouring 20ml of 0.5M hydrochloric acid into a beaker.
    - Adding a 3cm magnesium ribbon into the acid and start the stop watch immediately
    - Recording the time taken for magnesium to completely react
    - Repeating steps 1, 2 and 3 for all other concentrations of dilute hydrochloric acid
    - Designing a table for the presentation of your results
    - Plotting a graph of rate of reaction against concentration of HCl
    - Analysing and interpreting the graph, and make conclusions
    - Making presentations to the class

Assessment Guide:

- Check:
    - learner participation during the activity
    - correct handling of apparatus and chemicals
    - for correct plotting of the graph
    - for correct analysis and conclusions

Key Learning Points:

- Concentration is directly proportional to rate of chemical reaction. The graph below shows the relationship between rate of a chemical reaction and factors such as concentration, temperature and pressure.



Skills Developed:

- Communicating
- Collaborating
- Analysing

- Classifying
- Identifying

Activity 6.0: Exploring methods of controlling the rates of chemical reactions
In this activity, learners will explore methods of controlling rates of chemical reactions by research.

HOOK: A nurse on duty receives a patient with severe indigestion. The patient's stomach acid is reacting too slow with a standard antacid tablet, causing pain. The nurse crushes the tablet into powder, makes a highly concentrated solution, and adds a catalyst-like fizzing agent. The patient gets relief in seconds.

The quick relief experienced by the patient is as a result of rates of reactions.

Let's explore....

Suggested Teaching and Learning Materials:
- **Natural Materials:**
- **Artificial Materials**: Digital devices connected to internet, text books

Learning Environment Set-up:
- **Natural Environment:** School surroundings
- **Artificial Environment:** Classroom/Chemistry laboratory

Teacher's Roles:
- Introduce the activity using the hook
- Put learners in small groups
- Provide the materials to learners
- Ask leaners to:
    - research on different methods of controlling rates of chemical reactions domestically and in industry.
    - Justify each of the methods researched
    - present their findings

Learner's Activities:
- Responding to the hook
- Researching on different methods of controlling rates of chemical reactions domestically and in industry.
- Justifying each of the methods researched

- Presenting their findings

Assessment Guide:

- Check for:
    - learner participation during the activity
    - correctness of information researched

Key Learning Points:

The following are some of the methods of controlling the rates of chemical reactions in real life:

- By increasing or lowering the temperature for example,
    - lowering temperature helps prevent spoilage of food (refrigeration)
    - increasing the temperature in the oven increases the baking process
- By using of catalysts. For example, iron is used as a catalyst in Haber process to maximise the yields of ammonia
- Increasing pressure: For example, in car engines, fuel and oxygen are compressed to increase the speed of combustion
- By increasing or reducing the concentration of reactants. For example, diluting acids or bases to manage their reactivity
- Increasing or reducing or surface area. For example, powdered sugar dissolves faster in water than sugar cubes

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standard:

- Understanding of rates of chemical reactions demonstrated accordingly
- Factors affecting the rates of chemical reactions investigated accordingly

TOPIC 2.4.: REDOX REACTIONS

Introduction:

Redox reactions are chemical processes in which electrons are transferred from one substance to another, causing changes in oxidation states. These reactions are in two types; reduction and oxidation. These reactions are part of our everyday life. They occur when iron rusts, when batteries generate electricity, when fuels burn in car engines, and even when our bodies release energy from food during respiration. For example, the energy that powers our body cells is produced through redox processes in cellular respiration, similar in principle to how a battery works. Understanding redox reactions helps us explain corrosion, prevent food spoilage, design efficient fuels, and develop technologies such as electrochemical cells. Learning this topic therefore equips learners with knowledge that connects chemistry to real-world applications in energy production, environmental protection, health and industry.

General Competences:

- Collaboration
- Communication
- Analytical Thinking
- Environmental Sustainability
- Problem Solving

Sub-Topic 2.4.1: Oxidation and Reduction

Introduction:

Reduction and oxidation are chemical processes that occur together in many reactions and involve the transfer of electrons between substances. In daily life, oxidation takes place when metals rust, fuels burn, or food spoils, while reduction occurs in processes such as the extraction of metals from ores and the functioning of batteries. These reactions are also essential in our everyday lives. Understanding reduction and oxidation is important because they explain how energy is produced, how materials change, and how many industrial and biological processes occur, making them fundamental concepts in Chemistry and everyday life.

Specific Competence:

2.4.1.1 Interpret Redox Reactions

Key Terms:

- **Oxidation:** The process of gaining oxygen or losing hydrogen or losing electrons, or an increase in oxidation state
- **Reduction:** The process of losing oxygen or gaining hydrogen or gaining electrons, or a decrease in oxidation state

Learning Activities:

Activity 1.0: Analysing reduction and oxidation

Activity 1.1: Investigating reduction and oxidation through oxygen and hydrogen exchange

This activity will involve learners conducting an experiment of a reaction between copper (II) oxide and hydrogen. After observing the reaction, learners will write the equation for the reaction and discuss what happened to oxygen and hydrogen during the reaction.

HOOK: Imagine you are part of a team of young scientists working in a laboratory that studies metal corrosion. One morning, your supervisor places two metal strips on the table: one is shiny and new, while the other is covered with a reddish-brown layer of rust.

On investigating, you discover that the rusty metal gained oxygen from the air. In another experiment, you observe a substance losing oxygen when heated with hydrogen gas, forming water.

The big question arises: What happens to substances when they gain oxygen and what happens when they lose oxygen or gain hydrogen?

Suggested Teaching and Learning Materials:

- **Natural Materials**:
- **Artificial Materials**: Copper (II) oxide powder, Hydrogen gas source (or hydrogen generated from reaction between zinc and dilute hydrochloric acid), test tube and test-tube holder, Bunsen burner or spirit lamp, delivery tube and rubber stopper, heatproof mat, safety goggles and gloves, tongs, work sheet with activity procedure

Learning environment setup:

- **Natural Environment**: School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups

- Guide learners in setting up apparatus and conducting the experiment safely following activity procedure provided
- Ensure laboratory safety procedures are followed
- Ask probing questions such as:
    - Which substance lost oxygen?
    - Which substance gained hydrogen?
    - How can you tell reduction has occurred?
- Facilitate discussion and clarify misconceptions
- Ask learners to link observations to real-life examples (such as rusting, extraction of metals)
- Ask each group to present findings

Activity Procedure

1. Set up the apparatus to pass hydrogen gas over heated copper (II) oxide.
2. Gently heat the copper (II) oxide while hydrogen gas flows over it.
3. Observe and record changes in colour and formation of any liquid.
4. Write the balanced chemical equation for the reaction.

Learners' Activities:

- Setting up apparatus and carrying out the experiment safely
- Observing, recording, and interpreting results
- Writing balanced chemical equations
- Analysing which substance gained oxygen, which lost oxygen, which gained hydrogen, and which lost hydrogen
- Identifying oxidation and reduction in terms of oxygen/hydrogen exchange
- Presenting and justifying conclusions
- Reflecting on how the reaction demonstrates redox processes

Expected Reaction:

Copper (II) Oxide + Hydrogen gas → Copper + Water

$CuO + H_2 \rightarrow Cu + H_2O$

Assessment Guide:

- Check learners' participation and teamwork
- Oral questioning during the activity
- Review of balanced chemical equations
- Teacher can give learners homework questions such as:

1. In the reaction $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
   a) Which substance is oxidised? Explain.
   b) Which substance is reduced? Explain.
2. Give one everyday example of oxidation and one example of reduction.
3. Explain why oxidation and reduction must occur together.

Key Learning Points:

- **Oxidation** is the gain of oxygen or loss of hydrogen.
- **Reduction** is the loss of oxygen or gain of hydrogen.
- Oxidation and reduction occur simultaneously in a reaction (redox reaction).
- In the reaction, $CuO + H_2 \rightarrow Cu + H_2O$:
  - Copper (II) oxide is **reduced** (loses oxygen).
  - Hydrogen is **oxidised** (gains oxygen).
- These processes are important in metal extraction, fuel combustion, and energy production.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 1.2: Investigating reduction and oxidation through electron transfer

In this activity, learners will engage in a practical experiment to investigate reduction and oxidation through electron transfer.

During a laboratory experiment, learners placed a metal strip into a solution and notice that a new metal begins to form on its surface. Their teacher explains that tiny particles called electrons are being transferred between substances. Curious to understand the change, the students begin investigating how losing and gaining electrons leads to oxidation and reduction.

Get ready to investigate too.

Suggested Teaching and Learning Materials:

- **Natural Materials:**

- **Artificial Materials**: Zinc granules/zinc strip, copper (II) sulphate solution, beakers, sand paper (to clean zinc strip), filter paper, safety goggles and gloves, worksheet with activity procedure

Learning Environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Facilitate safe handling of chemicals
- Guide learners in observing carefully and recording results
- Ask probing questions such as:
    - Where did the electrons come from?
    - How does this relate to batteries?
- Support learners in writing correct half-equations
- Clarify misconceptions about oxidation and reduction

Activity Procedure

1. Place zinc granules into a beaker containing copper (II) sulphate solution.
2. Observe and record changes (e.g. colour change, copper deposits, temperature change).
3. Write the balanced chemical equation
4. Break the reaction into half-equations

Learners' Activities:

- Conducting the experiment safely and responsibly
- Observing, recording, and interpreting changes
- Writing balanced chemical equations and half-equations
- Analysing the chemical reactions in terms of electron loss and gain
- Collaborating, discussing and presenting findings clearly
- Relating the concept to real-life applications such as electrochemical cells
- Identifying which substance:
    - loses electrons
    - gains electrons
    - is oxidised
    - is reduced
- Presenting explanations using electron transfer diagrams

Assessment Guide:

- Check learner participation and teamwork
- Check for correct chemical equations, half-equations and explanations
- Teacher can give summative assessment questions such as:
    1. In the reaction between Iron and copper (II) chloride:
        a) Which substance is oxidised? Explain.
        b) Which substance is reduced? Explain.

2. Write the half-equations for the reaction.
3. Explain how this reaction demonstrates the working principle of a battery.

Key Learning Points:

- Reduction is the gain of electrons while Oxidation is the loss of electrons
- In the reaction between zinc and copper (II) sulphate, zinc is oxidised by losing electrons while copper (II) ions are reduced by gaining electrons
  $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
- The half reactions showing reduction and oxidation are as shown below:
  - $Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation)
  - $Cu^{2+} + 2e^- \rightarrow Cu$ (reduction)

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 1.3: Analysing redox reactions using oxidation state changes

In this activity, learners will analyse different reactions and identify elements whose oxidation states changed before and after the reaction. Learners will work in small groups under the guidance of the teacher.

HOOK: While helping at home, a learner notices that a cut apple slowly turns brown after being left in the open air. Later in a Chemistry class, the teacher explains that this everyday change because substances in the apple undergo a redox reaction.

Curious, the learner begin to investigate which atoms increased or decreased their oxidation states to determine which substance was oxidized and which was reduced. This leads them to analyze redox reactions using oxidation state changes

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Chart with rules for assigning oxidation numbers, Periodic Tables (one per group), markers and chart paper, worksheets with guided questions, calculators (optional), flash cards containing different redox equations for analysis.

Sample equations:

i. $2Mg + O_2 \rightarrow 2MgO$

ii. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

iii. $Cl_2 + 2KI \rightarrow 2KCl + I_2$

Learning Environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/ Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Review rules for assigning oxidation numbers
- Guide learners in applying rules correctly
- Ask questions such as:
  - what does an increase in oxidation number indicate?
  - why must oxidation and reduction occur simultaneously?
  - how does this apply to metal extraction or corrosion?
- Facilitate discussion and provide corrective feedback.
- Ask learners to link the lesson to real-life processes like rusting and industrial reduction of ores
- Ask selected groups to make presentations

Learners' Activities:

- Assigning oxidation numbers accurately
- Analysing changes in oxidation states
- Identifying oxidising and reducing agents
- Presenting logical explanations supported by calculations
- Engaging in peer discussion and critical questioning

Assessment Guide:

- Check
  - learner participation and reasoning
  - correctness of assigned oxidation numbers
- Teacher can give a class exercise or homework with questions such as:
  1. In the reaction, $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$: use oxidation states to identify the substance that is
     a) oxidised.
     b) reduced.

2. Explain one real-life application where oxidation state changes are important.

Key Learning Points:

- Oxidation is an increase in oxidation number while Reduction is a decrease in oxidation number
- Oxidation and reduction always occur together in a redox reaction
- Oxidation state analysis provides a systematic way to identify redox processes in complex reactions and is widely used in industrial chemistry, corrosion studies, and electrochemistry

Activity 2.0: Examining a redox reaction

In this activity, learners will observe a displacement reaction between zinc and copper (II) sulphate solution to identify how oxidation and reduction occur together.

HOOK: A copper plating factory uses a solution of copper salts to coat cheaper metals with a shiny layer of copper. This relates to a classroom experiment where an iron nail in copper sulphate solution develops a reddish coating.

By examining which substances lose electrons and which gain electrons, you can see how redox reactions are applied in metal plating, corrosion prevention, and everyday products.

Let's examine!

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Zinc granules or strips, copper (II) sulphate solution, beakers/ test-tubes, stirring rods, sandpapers, safety goggles and gloves

Learning Environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide the materials to learners
- Review safety precautions (handling chemicals, no ingestion, proper disposal)
- Demonstrate briefly how to clean zinc with sand paper before use
- Guide learners to:
    - make and record observations

    - write the balanced chemical equation
    - examine the chemical equation and interpret reduction and oxidation
- Ask guiding questions:
    - what evidence shows a chemical reaction occurred?
    - which substance is oxidised? Explain.
    - which substance is reduced? Explain.
    - why must oxidation and reduction occur together?
- Facilitate discussion linking observations to electron transfer
- Ask learners to make presentations

Learners' Activities:
- Pouring copper (II) sulphate solution into a beaker
- Adding cleaned zinc strip to the solution
- Observing and recording observations
- Writing the balanced equation
- Examining the chemical equation and interpreting reduction and oxidation
- Providing responses to questions
- Making presentations

Assessment Guide:
- Check learner participation and laboratory skills
- Give questions such as:
    1. Identify the oxidized substance in the reaction.
    2. Identify the reduced substance.
    3. Write half-reactions for oxidation and reduction.
    4. Explain why oxidation and reduction must occur simultaneously.

Extension of the Activity

Learners to research how redox reactions occur in:

i. Rusting of iron
ii. Dry cells and car batteries
iii. Extraction of metals from ores

They prepare a short report or poster explaining the redox process involved.

Key Learning Point:
- Redox reactions are chemical reactions in which oxidation and reduction occur simultaneously. Oxidation and reduction occur simultaneously in chemical reactions

because the two processes involve electron transfer. Therefore, the electron(s) lost by one substance should be gained by another substance.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Activity 3.0: Identifying characteristics of oxidizing and reducing agents

There are two sub-activities under this activity to help learners identify characteristics of oxidising and reducing agents.

Activity 3.1: Identifying characteristics of oxidising agents using potassium iodide

In this activity, learners will use potassium iodide solution and starch solution to test different substances and identify which ones act as oxidising agents based on the observed colour change.

HOOK: While cooking at home, a learner notices that bleach can turn things white quickly. Curious about how it works, the student goes in the laboratory and tests different substances with potassium iodide solution and observes the solutions as they change colour.

By observing which substances cause the iodide to lose electrons, learner learn to identify oxidizing agents and understand their role in everyday life, from disinfecting surfaces to bleaching clothes.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Potassium iodide (KI) solution, starch solution, droppers, test tubes, test tube rack, white tile, samples of possible oxidising agents (e.g. acidified potassium permanganate, chlorine water, hydrogen peroxide, dilute nitric acid), distilled water (as a control)

Learning Environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity by reviewing oxidation and reduction concepts
- Put learners in small groups
- Review laboratory safety procedures
- Emphasize observation skills
- Ask learners to conduct the experiment following the activity procedure
- Ask learners to make and record observations
- Assist groups experiencing difficulties
- Ask learners to interpret the observations and results

Safety tip:

- Ensure learners wear goggles and laboratory coats
- Ensure safe disposal of chemicals

Activity Procedure:

Using potassium iodide solution and starch solution

1. Place $2cm^3$ of potassium iodide solution into a test tube.
2. Add a few drops of the analyte solution (e.g. potassium permanganate).
3. Add a few drops of starch solution.
4. Shake gently.
5. Observe and record any colour change.

Expected Observation:

- Formation of a blue-black colour indicating iodine formation. Therefore, the substance is an oxidising agent
- When an oxidising agent is present, it oxidises iodide ions (I^-) to iodine (I_2). The iodine produced reacts with starch to form a characteristic blue-black colour. This colour change provides clear visual evidence that oxidation has occurred.

Learners' Activities:

- Following experimental procedures carefully
- Making and recording observations in terms of initial and final colours
- Making inferences
- Participating in group discussion
- Drawing conclusions from results

Assessment Guide:

- Check learner participation, correct handling of apparatus, observations, interpretation of results and correctness of conclusions made

- Teacher can ask questions such as:
    - Why does potassium iodide act as a reducing agent?
    - What causes the blue-black colour?
    - What does the colour change indicate?
- Monitor if learners are handling chemicals safely
- Check if learners are recording observations accurately

Key Learning Points:

- Oxidising agents are substances that cause other substances to lose electrons while they gain electrons. In the process, they are reduced
- One way to identify an oxidising agent is by using potassium iodide (KI) solution or acidified potassium iodide paper containing starch
- When an oxidising agent reacts with potassium iodide (KI), iodide ions (I^-) are oxidised to iodine (I_2) which forms a blue-black colour in the presence of starch

Reaction: $2I^- \rightarrow I_2 + 2e^-$

Activity 3.2: Identifying characteristics of reducing agents

In this activity learners will identify reducing agents using acidified potassium dichromate and acidified potassium permanganate as oxidising agents. The activity will be a practical experiment involving learners working in small groups.

HOOK: While making homemade lemonade, a learner adds few slices of lemon to water. The learner accidentally drops a vitamin C tablet in the mixture and notices that the drink does not turn brown.
Curious, the learner investigates in the lab how certain substances donate electrons to other chemicals. This experiment helps the learner to identify reducing agents and understand their role in everyday life, such as preserving food and preventing oxidation.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Acidified potassium dichromate solution (orange), acidified potassium permanganate solution (purple), dilute sulfuric acid, test tubes and droppers, test tube rack, white tile, samples of possible reducing agents (e.g., iron (II) sulphate solution, sodium sulphite solution, ethanol), distilled water (control), safety goggles and lab coats

Learning Environment setup:

- **Natural Environment:** School surroundings

- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Ensure safety precautions are in place (goggles, proper waste disposal)
- Emphasise careful observation
- Ask learners to:
    - conduct the experiment following the activity procedure
    - make and record observations
    - interpret the observations and results
- Assist groups experiencing difficulties

Safety tip: Ensure proper disposal of chromium-containing waste

Activity Procedure

Part A: Using acidified potassium dichromate

1. Place 2cm³ of acidified potassium dichromate solution in a test tube.
2. Add a few drops of the analyte solution (e.g. iron (II) sulphate solution)
3. Shake gently.
4. Observe and record any colour change.

Expected Observation:

- Orange → Green; Indicates the analyte is a reducing agent

Part B: Using acidified potassium permanganate

1. Place 2cm³ of acidified potassium permanganate solution in a test tube.
2. Add a few drops of the analyte solution (e.g. ethanol)
3. Shake gently.
4. Observe and record any colour change.

Expected Observation

- Purple → Colourless (or pale pink); Indicates the analyte is a reducing agent

Learners' Activities:

- Following experimental procedures carefully
- Making and recording observations in terms of initial and final colours
- Making inferences
- Participating in group discussion
- Drawing conclusions from results

Assessment Guide:

- Check for collaboration among learners, correct handling of apparatus, correct observations and interpretation of results.
- Check if learners are recording observations accurately

Key Learning Points:

- A reducing agent is a substance that donates electrons and causes another substance to be reduced. It is oxidised in the process.
- When using acidified potassium dichromate ($K_2Cr_2O_7$), the orange solution turns green. The dichromate ions are reduced to chromium ions ($Cr_2O_7^{2-} \rightarrow Cr^{3+}$). The colour change indicates the presence of a reducing agent.
- When using acidified potassium permanganate ($KMnO_4$), the purple solution turns colourless (or very pale pink). The manganate ions are reduced to manganese ions ($MnO_4^{-} \rightarrow Mn^{2+}$). The colour change indicates the presence of a reducing agent.

Activity 4.0: Determining oxidation numbers of elements with variable oxidation states

In this activity, learners will work collaboratively in groups to determine oxidation numbers of elements that exhibit variable oxidation states.

HOOK: At home, a learner notices that rust forms faster on some iron tools than others. The learner asks the teacher who explains that some metals change their charge in different reactions.

In the lab, the learner practice determining oxidation numbers of elements with variable states to understand why other iron tools rust faster than others.

Investigate.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Worksheet with chemical formulae of compounds (e.g. $FeCl_2$ and $FeCl_3$, $KMnO_4$ and MnO_2, H_2SO_4 and H_2SO_3, Cu_2O and CuO, SO_2 and SO_3), chart with rules for assigning oxidation numbers, Periodic Tables

Learning Environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide the materials to the learners
- Ask learners to determine the oxidation numbers of the underlined elements in each pair of compounds on the work sheet:

    a) Showing full working
    b) Identifying whether the element is in a higher or lower oxidation state compared to another compound on the sheet.
- Ask learners to make presentations
- Correct misconceptions immediately
- Reinforce systematic problem-solving

Learners' Activities:

- Determining the oxidation numbers of the underlined elements in each pair of compounds on the work sheet showing full working
- Identifying whether the element is in a higher or lower oxidation state compared to another compound on the sheet
- Making presentations

Assessment Guide:

- Check learner participation and accuracy of calculations
- Check for correct application of rules and logical working steps

Key Learning Points:

- Oxidation numbers are used to show the oxidation state of an element in a compound
- Oxidation number (oxidation state) is the apparent charge an atom would have if all bonds were ionic
- Some rules for assigning oxidation numbers are as follows:
    i. Oxidation number of an element in its free state is 0
    ii. Oxidation number of Group I elements is +1
    iii. Oxidation number of Group II elements is +2
    iv. Oxidation number of oxygen is -2, except in peroxides
    v. Oxidation number of hydrogen is +1, except in metal hydrides
    vi. Oxidation number of simple ions is equal to the charge on the ion
    vii. The sum of oxidation numbers for all the elements in a neutral compound is 0

viii. The sum of oxidation numbers of all elements in a polyatomic ion is equal to the charge on the ion

- Examples of elements with variable oxidation states include the following:
    - Iron: +2, +3
    - Copper: +1, +2
    - Nitrogen: –3 to +5
    - Sulphur: –2 to +6
    - Chlorine: –1 to +7
    - Manganese: +2 and +7
    - Chromium: +3 and +6

Activity 5.0: Deducing a redox reaction using oxidation numbers

In this activity, learners will act as chemical detectives investigating whether a reaction is redox and identifying the agents involved. They will deduce oxidation and reduction in using oxidation numbers.

HOOK: You could have noticed that when you add lemon to tea, the colour of the tea changes over time. You can understand the Chemistry behind this by investigating and deducing redox reaction, identifying which substance is oxidized or reduced. You can deduce redox reactions using oxidation numbers.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Reaction cards (printed on small cards), Periodic Tables, oxidation number rule sheet, flip chart/manila paper, markers, sticky notes, reaction cards with the following equations:
    1. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
    2. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
    3. $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
    4. $2Mg + O_2 \rightarrow 2MgO$
    5. $Cl_2 + 2KI \rightarrow 2KCl + I_2$
    6. $HCl + NaOH \rightarrow NaCl + H_2O$

Learning Environment Setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide the materials to learners

- Assign a specific reaction on the work card to each group
- Ask learners to:
    - assign oxidation numbers to all elements in the reaction
    - identify any changes in oxidation numbers
    - decide whether the reaction is redox or not
    - identify the oxidising agent and reducing agent
    - make presentations
- Consolidate learner presentations

Learners' Activities:

- Assigning oxidation numbers to all elements in the reaction
- Identifying any changes in oxidation numbers
- Deciding whether the reaction is redox or not
- Identifying the oxidising agent and reducing agent
- Making presentations and defend conclusions

Assessment Guide:

- Check
    - correct assignment of oxidation numbers
    - logical reasoning
    - learner participation
    - ability to justify answers

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Learning Activity 6.0: Describing a non-redox reaction

In this activity, learners will engage in research to describe non-redox reactions.

HOOK: While working in the kitchen, a learner pours vinegar (CH_3COOH) over baking soda ($NaHCO_3$) and watches it fizzing vigorously. Having learnt about redox reactions, the learner suspects that a redox reaction was taking place. The learner writes the equation for the reaction and uses oxidation numbers to check if the reaction is redox. The learner concludes that it is not a redox reaction.

Is the learner correct or not? Let's find out.

Suggested Teaching and Learning Materials:

- **Natural Materials:**
- **Artificial Materials**: Digital devices connected to internet, text books, flip charts, markers

Learning environment setup:

- **Natural Environment:** School surroundings
- **Artificial Environment**: Classroom/Chemistry or computer laboratory

Teacher's Roles:

- Introduce the activity
- Put learners in small groups
- Provide the materials to learners
- Ask learners to use available sources of information to:
    - Search on non-redox reactions
    - Write examples on non-redox reactions
    - Use knowledge on oxidation and reduction to justify why the reactions are non-redox
    - Make presentations
- Consolidate learner presentations

Learners' Activities:

- Searching on non-redox reactions
- Writing examples on non-redox reactions
- Justifying why the reactions are non-redox using knowledge on oxidation and reduction
- Making presentations

Assessment Guide:

- Check the correctness of the researched information and justifications given

Key Learning Points:

A non-redox reaction is a chemical reaction in which there is no:

- change in oxidation numbers of elements
- transfer of electrons

Common examples include:

- Neutralisation reactions, for example: $HCl + NaOH \rightarrow NaCl + H_2O$
- Double displacement reactions, for example, $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

In these reactions oxidation numbers remain unchanged before and after the reaction.

Skills Developed:

- Communicating
- Collaborating
- Analysing
- Classifying
- Identifying

Expected Standard:

- Redox reactions interpreted correctly

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